

FAITHFUL EXPLANATIONS OF BLACK-BOX NLP MODELS USING LLM-GENERATED COUNTERFACTUALS

Yair Ori Gat^{T*}, Nitay Calderon^{T*}, Amir Feder^C,
Alexander Chapanin^T, Amit Sharma^M and Roi Reichart^T

^TFaculty of Data and Decision Sciences, Technion, IIT

^CColumbia University Data Science Institute, ^MMicrosoft Research India

*Equal contribution. Corresponding author: nitay@campus.technion.ac.il

ABSTRACT

Causal explanations of the predictions of NLP systems are essential to ensure safety and establish trust. Yet, existing methods often fall short of explaining model predictions effectively or efficiently and are often model-specific. In this paper, we address model-agnostic explanations, proposing two approaches for counterfactual (CF) approximation. The first approach is CF generation, where a large language model (LLM) is prompted to change a specific text concept while keeping confounding concepts unchanged. While this approach is demonstrated to be very effective, applying LLM at inference-time is costly. We hence present a second approach based on matching, and propose a method that is guided by an LLM at training-time and learns a dedicated embedding space. This space is faithful to a given causal graph and effectively serves to identify matches that approximate CFs. After showing theoretically that approximating CFs is required in order to construct faithful explanations, we benchmark our approaches and explain several models, including LLMs with billions of parameters. Our empirical results demonstrate the excellent performance of CF generation models as model-agnostic explainers. Moreover, our matching approach, which requires far less test-time resources, also provides effective explanations, surpassing many baselines. We also find that Top-K techniques universally improve every tested method. Finally, we showcase the potential of LLMs in constructing new benchmarks for model explanation and subsequently validate our conclusions. Our work illuminates new pathways for efficient and accurate approaches to interpreting NLP systems.

1 INTRODUCTION

Providing faithful explanations for Natural Language Processing (NLP) model predictions is imperative to guarantee safe deployment, establish trust, and foster scientific discoveries (Amodei et al., 2016; Goodman & Flaxman, 2017; Guidotti et al., 2019; Jacovi & Goldberg, 2020). These aspects are particularly significant in NLP, where the complexity of language and the opaque behavior of black-box models. Many past works focus on *what knowledge a model encodes* (Lyu et al., 2022). However, just because a model encodes a myriad of features does not mean all are utilized in decision-making. For an explanation to be genuinely faithful and accurately depict a model’s underlying reasoning, it is crucial to establish causality. Recognizing this inherent link and following previous works (Vig et al., 2020; Geiger et al., 2020; Feder et al., 2021c), this paper introduces a theoretical framework that binds the two together, providing another evidence that non-causal explanation methods can occasionally fall short of being truly faithful.

In contrast to model explanation techniques that *often conflate correlation with causation*, causal-inspired methods often contrast predictions for an input example with those of its counterfactual (Soulos et al., 2020; Elazar et al., 2021; Finlayson et al., 2021). Indeed, counterfactuals are at the highest level of Pearl’s causal hierarchy (Pearl, 2009), highlighting how changes lead to a different prediction. However, they cannot be acquired without knowing the complete data-generating process (or structural model) of the text (Balke & Pearl, 1994), which is not practical for a given real-world problem. Hence, we turn to *counterfactual approximations* (CFs): imagining how a given text would

look like if a certain concept in its generative process were different (e.g., a different domain for an Amazon product review, as in Calderon et al. (2022)). Nevertheless, acquiring CFs in past work was limited to simple local manipulations (Ribeiro et al., 2020; Ross et al., 2021; Wu et al., 2021) or costly manual annotation (Gardner et al., 2020; Kaushik et al., 2020a), hindering practical causal effect estimation of high-level concepts on NLP models.

To address the above limitations, we rely on Large Language Models (LLMs) and introduce two practical approaches for model explanation. The first approach is *Counterfactual Generation*, in which a generative model (an LLM) is prompted to change a concept of a text while holding confounding concepts fixed. The first contribution of this paper is our empirical evidence that *CF generating LLMs are SOTA explainers*. However, utilizing LLMs comes with a high computational and financial cost. Moreover, the slow autoregressive generation process may limit the application of CF generation methods, especially in scenarios demanding real-time explanations or when explaining vast quantities of data (Calderon et al., 2023). Therefore, we introduce an efficient alternative: *Matching*. In comparison to LLM-based CF generation, the matching technique is up to 1000 times faster (§B)

We use matching to estimate the causal effect, pairing each treated unit with one or more control units with similar observed characteristics, i.e., identifying CF approximations within a pre-defined candidate set. To enhance matching quality, we propose in §3.4 a novel method for learning a causal embedding space. Given a causal graph (e.g., table 1), we first employ an LLM (ChatGPT) for generating CFs (note that the LLM is used only during the learning phase). We also identify possible valid matches: Instances with identical values in observable adjustment variables (that satisfy the back-door criterion). Subsequently, we train our causal representation model by minimizing an objective composed of contrastive loss components. The causal representation model learns to encode examples similarly to their LLM-generated CFs or matched examples. Conversely, it is trained to produce dissimilar representations for the query example and any misspecified CFs. As a result, representations remain faithful to the causal graph and can effectively identify matching candidates.

In §3.1, we propose a simple, intuitive, and essential criterion for explanation methods: *Order-faithfulness* – If the explanation method ranks one concept’s impact above another’s, its true causal effect should genuinely be greater. We then theoretically show that approximating CF methods are always order-faithful, in contrast to non-causal explanation methods (that overlook confounders, for example). Moreover, since CFs do not depend on the explained model, they enable causal estimation in a *model-agnostic manner*. Model-agnostic explanation methods offer numerous advantages, especially during model selection, debugging, and deployment. For example, developers juggling multiple models can effortlessly rank them based on their vulnerability to confounding biases (such as gender bias). Hence, we focus only on model-agnostic explanations in this study.

We undertake rigorous experiments to compare our causal language representation method against various generative and matching baselines. For this purpose, we employ CEBaB (Abraham et al., 2022), the only established benchmark for evaluating causal explanation methods. We estimate the causal effect of 24 concept interventions on the predictions of five models, three small fine-tuned LMs, and two zero-shot LLMs with billions of parameters. In §5.1 we also conduct an ablation study to examine the impact of each design choice within our causal model.

Our empirical results reveal three key findings: (1) The counterfactual generation approach provides strong black-box model explanations. (2) Our method for learning causal representations for matching outperforms all the matching baselines, including the best-performing method from the CEBaB paper (see Figure 6). (3) Top- K matching universally greatly enhances the explanatory capabilities of all examined methods, including generative models, when generating multiple CFs.

Finally, building on our findings that LLMs can produce high-quality CFs, we address a pressing challenge: The lack of benchmarks for NLP model explanation methods. In §6 we demonstrate how to construct such a benchmark by leveraging GPT-4 to generate pairs of inputs and their CFs (used as “golden CF”). We introduce a new CEBaB-like dataset, focusing on Stance Detection, an NLP task that determines the position expressed by the speaker towards a particular topic (e.g., abortions, climate change) (Küçük & Can, 2021; Viehmann et al., 2023), where model explanations may guide decision-making processes or reveal societal biases. Our novel dataset allows evaluation in an out-of-distribution setting, such as when the matching candidate set has topics differing from the test examples. Utilizing the new dataset, we observe similar patterns to our main conclusions. We hope that this study will pave the way for safer, more transparent, and more accountable AI systems.

2 RELATED WORK

Explanation methods and causality. Probing is a common technique for understanding what model internal representations encode. In probing, a small supervised (Conneau et al., 2018; Tenney et al., 2019) or unsupervised (Clark et al., 2019; Manning et al., 2020; Saphra & Lopez, 2019) model is used to measure whether specific concepts are encoded at specific places in a network. While probes have helped illuminate what models have learned from data, Geiger et al. (2021) show that probes cannot reliably provide causal explanations. Feature importance tools can also be seen as explanation methods (Molnar, 2020), often restricted to input features, although gradient-based methods can quantify the relative importance of hidden states as well (Zeiler & Fergus, 2014; Binder et al., 2016; Shrikumar et al., 2017; Gat et al., 2022). The Integrated Gradients method of Sundararajan et al. (2017) has a natural causal interpretation stemming from its exploration of baseline (CF) inputs (Geiger et al., 2021). However, even where these methods can focus on internal states, it remains challenging to connect their analyses with real-world concepts that do not reduce to simple properties of inputs. In this study, we focus on the explanation of high-level real-world concepts.

Intervention-based methods involve modifying inputs or internal representations and studying the effects that this has on model behavior (Lundberg & Lee, 2017; Ribeiro et al., 2016). Recent methods **perturb** input or hidden representations to create CF states that can then be used to estimate causal effects (Elazar et al., 2021; Finlayson et al., 2021; Soulos et al., 2020; Vig et al., 2020; Geiger et al., 2021; Feder et al., 2021b;a). However, these methods are prone to generating implausible inputs or network states unless the interventions are carefully controlled (Geiger et al., 2020). Recently, Wu et al. (2023) proposed the Causal Proxy Model, a novel explanation method inspired by S-learner (Künzel et al., 2019), which mimics the behavior of the explained model when the input is a CF. Although CPM is effective, the need to train a distinct explainer for each explained model is a major disadvantage. Conversely, our study focuses on model-agnostic explanation methods.

Approximating counterfactuals. A common use-case for CF examples in machine learning is for *data augmentation*. These CFs involve perturbations to confounding factors (Garg et al., 2019), or to the label (Kaushik et al., 2019; 2020b; Jha et al., 2020). CF examples can be generated through manual editing, heuristic keyword replacement, or automated text rewriting (Kaushik et al., 2019; Gardner et al., 2020; Shekhar et al., 2017; Garg et al., 2019; Feder et al., 2021b; Zmigrod et al., 2019; Riley et al., 2020; Wu et al., 2021; Mao et al., 2021; Rosenberg et al., 2021). Manual editing is accurate but expensive, while keyword-based methods can be limited in coverage and difficult to generalize across languages (Antoniak & Mimno, 2021). Generative approaches offer a balance of fluency and coverage (Zhou & Wu, 2023). CF examples help address causal inference’s missing data issues, but generating meaningful CFs without LLMs is challenging. Our work uses causal representations to match examples from the treatment group (i.e., the concept for which the causal model effect is estimated) with similar examples from the control group.

Faithful Explanations. Faithfulness is a desirable property of any explanation method, broadly defined as *the ability of the method to provide accurate descriptions of the underlying reasoning of the explained model* (Jain & Wallace, 2019; Jacovi & Goldberg, 2020). As of yet, there is no universally accepted or formal definition of this term (Lyu et al., 2022). Instead, many opt to characterize faithfulness using a variety of evaluation metrics, including Sensitivity (Sundararajan et al., 2017; Adebayo et al., 2018; Kindermans et al., 2018), Consistency (Liu et al., 2022), Feature Importance Agreement (Wiegrefe et al., 2021), Robustness (Ghorbani et al., 2019; Yin et al., 2022) and Simulatability (Sushil et al., 2018; Sia et al., 2023; Li et al., 2020; Pruthi et al., 2022). Most of these works focus on feature-level explanations, rely on access to the internal mechanism of the model, train an external model, and occasionally conflate between correlation and causation. In this work, we focus on high-level concept explanations. Relying on the causality literature, in §3.1 we propose a simple, intuitive formal criterion, which is also a necessary condition for the above broad definition of faithfulness: *Order-Faithfulness*.

3 METHOD

3.1 MODEL EXPLANATION WITH COUNTERFACTUALS

This study focuses on black-box NLP model explanations by estimating the causal effect of high-level concepts on model prediction. The first requirement for any causal estimation method is access to a

causal graph that describes our causal beliefs, i.e., the concepts and the relationships between them (e.g., see the causal graph in Figure 1). Notice that we do not assume access to the complete *data-generating process (DGP)*, which besides the causal graph, also describes the exact mathematical model that quantitatively defines the relationships and interactions among concepts and the exogenous variables. Furthermore, the causal effect it must be identifiable from the graph (for an example of a causal graph that is non-identifiable, see Figure 4 in Cinelli et al. (2022)), and therefore, in this study, we discuss only identifiable causal graphs.

To provide an accurate explanation, we should estimate the *Average Treatment Effect (ATE)* (Pearl, 2009) or the *Individual Treatment Effect (ITE)* (Shpitser & Pearl, 2006; Shalit et al., 2017). Specifically, the treatment is a high-level concept influencing the text (such as *ambiance* in a restaurant review), and the outcome variable is a prediction of a text classifier. When discussing model explanation in the sense of the causal effect of a concept on the model prediction, the common versions of the ATE and the ITE are the *Causal Concept Effect (CaCE)* (Goyal et al., 2019) and the *Individual Causal Concept Effect (ICaCE)* (Abraham et al., 2022), which we next formally define.

Given a DGP $\mathcal{G}$, an intervention on a treatment variable $T : t \rightarrow t'$ (for simplicity we sometimes write $T \leftarrow t'$), a model f , and a query example $\mathbf{x}_t$, the CaCE and ICaCE are defined to be:

$$\begin{aligned} \text{CaCE}_f(T, t, t') &= \mathbb{E}_{\mathbf{x}' \sim \mathcal{G}} [f(\mathbf{x}') | do(T = t')] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{G}} [f(\mathbf{x}') | do(T = t)] \\ \text{ICaCE}_f(\mathbf{x}_t, T, t') &= \mathbb{E}_{\mathbf{x}' \sim \mathcal{G}} [f(\mathbf{x}') | \mathbf{x}_t, do(T = t')] - f(\mathbf{x}_t) \end{aligned} \quad (1)$$

One way to provide an estimation of the causal effect and explain the model is by using *counterfactuals (CFs)*, which enables causal estimation in a model-agnostic manner (because CFs do not depend on the explained model). While we are aware of the causal graph that encodes our causal beliefs, for a given problem we cannot access the complete DGP nor the variable values of an example, and therefore, we cannot produce gold CFs. Given this, we propose two approaches for approximating them: Counterfactual generation (§3.3) and Matching (§3.4).

If $\tilde{\mathbf{x}}_{t'}$ is an approximated CF of the text $\mathbf{x}_t$ resulting from the intervention $T : t \rightarrow t'$ (can be either a model-generated CF or a match), then the causal effect estimators are:

$$\begin{aligned} \widehat{\text{ICaCE}}_f(\mathbf{x}_t, T, t, t') &:= \widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}) = f(\tilde{\mathbf{x}}_{t'}) - f(\mathbf{x}_t) \\ \widehat{\text{CaCE}}_f(T, t, t') &= \frac{1}{|\mathbb{D}|} \left(\sum_{\mathbf{x}_t} \widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}) + \sum_{\mathbf{x}_{t'}} \widehat{\text{ICaCE}}_f(\tilde{\mathbf{x}}_t, \mathbf{x}_{t'}) + \sum_{\mathbf{x}} \widehat{\text{ICaCE}}_f(\tilde{\mathbf{x}}_t, \tilde{\mathbf{x}}_{t'}) \right) \end{aligned} \quad (2)$$

Additionally, we can provide a more robust estimator (see the third findings in our results §5) by performing *Top-K matching*:

$$\widehat{\text{ICaCE}}_f(\mathbf{x}_t, T, t, t') = \frac{1}{K} \sum_{i=1}^K \widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^i) \quad (3)$$

3.2 COUNTERFACTUALS AS AN IDEAL MODEL EXPLANATION

While definitions of faithfulness vary across literature, there is a consensus that a faithful explanation method should accurately describe a model’s underlying reasoning (Rudin, 2019; Jacovi & Goldberg, 2020; Dasgupta et al., 2022). Lyu et al. (2022) distinguish between two primary avenues of model explanations: *what knowledge a model encodes* and *why a model makes certain predictions*. Nevertheless, a prevailing misconception is that the *what* avenue directly informs *what the model uses in making predictions* — a flawed notion. While a model can encode a myriad of features, it does not imply that all are utilized in decision-making. Explanations derived from the *what* avenue, thus, tend to be correlational rather than causal. Focusing on the *why* is imperative for understanding a model’s reasoning mechanisms, making causality indispensable. Consequently, causality must be embedded at the core of the definition of faithfulness.

Therefore, relying on the causality literature, we propose a simple, intuitive, and essential criterion: *Order-faithfulness*. This implies that if the explanation method ranks the impact of one concept (or intervention) higher than another, the true causal effect of the first concept should genuinely exceed that of the second. Order-faithfulness is crucial in scenarios like model selection and deployment.

For instance, to determine the fairness of a hiring model, it is essential to know whether it incorrectly prioritizes non-relevant concepts (e.g., gender) over genuine concepts (such as experience). We next formally define this necessary property.

Definition (Order-Faithfulness). *Given an i.i.d. dataset $\mathbb{D}$ sampled from a DGP $\mathcal{G}$, an explained model f and a pair of interventions $C_1 : c_1 \rightarrow c'_1, C_2 : c_2 \rightarrow c'_2$, an explanation method S is **order-faithful** if:*

$$\mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S(f, C_1, c_1, c'_1)] > \mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S(f, C_2, c_2, c'_2)] \iff \text{CaCE}_f(C_1, c_1, c'_1) > \text{CaCE}_f(C_2, c_2, c'_2) \quad (4)$$

Note that the definition above does not mandate the explanation method to estimate the causal effect accurately or generate scores proportional to the causal effects. Instead, it simply requires the method to be rank-preserving. In addition, it holds for comparing the importance of two distinct concepts, as many attribution methods like LIME do (Ribeiro et al., 2016), and it also holds when assessing the importance of changes in the same concept. We believe that order-faithfulness is a necessary condition for the broad definition of faithfulness above: *to accurately describe the underlying reasoning of the explained model* (Jacovi & Goldberg, 2020; Lyu et al., 2022). If an explanation method ranks the importance of one concept above another, but in fact its causal effect is smaller, then the method does not accurately reflect the actual reasoning process of the model being explained.

In Appendix §A we provide detailed formal definitions of two explanation methods: Approximated CF explanation methods and Non-causal explanation methods (which we define as any function of the observed data). Furthermore, we prove the following theorem, which elucidates why approximated CF methods are always order-faithful, unlike non-causal ones.

Theorem. *For an explained model f , (1) The approximated CF explanation method S_{CF} is order-faithful for every DGP $\mathcal{G}$ and a pair of interventions; (2) For every DGP $\mathcal{G}$, there exist a DGP $\mathcal{G}'$ resulting from a small modification of $\mathcal{G}$, a model f' , and a pair of interventions $C_1 : c_1 \rightarrow c'_1, C_2 : c_2 \rightarrow c'_2$ such that the non-causal explanation method S_{NC} is not order-faithful.*

Proof sketch: We first prove that approximated CF methods are always faithful by showing the expected prediction of an approximated CF is equal to the interventional one (conditioned on the do operator). To prove the second part, we construct $\mathcal{G}'$ by introducing a new unobserved confounder variable (the “small modification”) that reverses the order of two interventions’ causal effect on a new model f' . In addition, we ensure that the non-causal method produces the same explanations as it produces for $\mathcal{G}$ since the generation process of $\mathbf{x}$ is the same in the two DGPs. Hence, S_{NC} cannot be order-faithful in $\mathcal{G}$ and $\mathcal{G}'$ at the same time.

The theorem underscores the importance of developing causal-inspired explanation methods and suggests why our CF-based approaches are more effective than other baselines.

3.3 LLM-GENERATED COUNTERFACTUALS

The first approach we introduce for CF approximation is *Counterfactual Generation*, in which a generative model (an LLM) is prompted to change an attribute conveyed in the text. We explicitly inject our causal beliefs into the prompt using the fundamental causal concept of *adjustment*. Given a causal graph and an intervention $T : t \rightarrow t'$, we first derive the adjustment set: Confounding concepts satisfy the back-door criterion (see Appendix §C). Complementary, the causal graph also implies which variables should not be adjusted: Mediators (variables that are part of the causal pathway) and Colliders (variables that are caused by both the treatment and the outcome). Adjusting these variables can lead to incorrect CFs. If we are interested in the direct causal effect (and not the total effect) of the treatment variable, we can add the mediators to the adjustment set.

Then, given a query example $\mathbf{x}_t$, we prompt the LLM to generate a CF. The prompt instructs the LLM to intervene in the text by changing the value of the concept T from t to t' . Additionally, we instruct the LLM to consider the adjusted concepts (confounders) and hold them fixed. Optionally, we can also specify the non-adjusted concepts (such as mediators and colliders) if they exist and ask the LLM to consider that the intervention on T might affect these concepts. This specification can increase the precision of the generated CF. For the Top-K technique, we simply generate multiple CFs.

Nevertheless, utilizing an LLM at inference time is frequently impractical because of its prolonged latency, high financial costs, or privacy concerns that prevent data from being sent to external servers

Table 1: Toy illustration of examples (and their concept values) sampled from the four sets of a given query example and an intervention of changing service to positive ($S \leftarrow +$). The misspecified CF ($\mathbb{X}_{\neg\text{CF}}$) resulted from a wrong intervention of removing the ambiance concept ($A \leftarrow ?$).

Query	Excellent lobster and decor, but rude waiter.
	$F : + \quad S : - \quad N : ? \quad A : +$
$\mathbb{X}_{\text{CF}}$	Excellent lobster and decor, and friendly waiter.
	$F : + \quad S : + \quad N : ? \quad A : +$
$\mathbb{X}_{\text{M}}$	A cozy atmosphere with great pizza and service.
	$F : + \quad S : + \quad N : ? \quad A : +$
$\mathbb{X}_{\neg\text{CF}}$	Excellent lobster and rude waiter.
	$F : + \quad S : + \quad N : ? \quad A : ?$
$\mathbb{X}_{\neg\text{M}}$	Too expensive for a loud chaos with bland food.
	$F : - \quad S : ? \quad N : - \quad A : -$

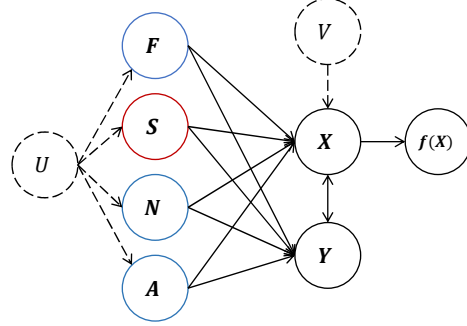


Figure 1: Our illustration of the causal graph of the CEBaB benchmark (Abraham et al., 2022). U and V are (unobserved) exogenous variables representing the state of the world. For example, V may account for style, syntax, and length. The four concepts F (food), S (service), A (ambiance) and N (noise) affect both the textual review X and its five-star rating Y . The relationship between X and Y can be causal or anti-causal. f is the model we wish to explain, which is trained to predict Y from X . In this study, we are interested in estimating the causal effect of changing a concept on the predictive model f (e.g., changing the value of the red variable S from positive to negative).

(see the discussion in Calderon et al. (2023) and our latency measurements in §B). Considering these challenges, in the next subsection (§3.4) we introduce an efficient alternative, the *matching* technique.

3.4 CAUSAL REPRESENTATION LEARNING FOR MATCHING

Although counterfactual generation is a valuable model explanation approach, employing LLMs during inference can be infeasible. Conversely, there is an opportunity to harness model-generated CFs to train an efficient method that seeks approximations within a pre-defined set of candidate texts, mitigating the need for direct generation. This approach is known as *matching* (Stuart, 2010) and endeavors to match each treatment unit with its most similar control unit. To allow effective matching, we introduce a novel method that learns to encode texts in a space faithful to the causal graph, i.e., each treatment unit resonates closely with its CF counterparts and remains distinct from confounding texts. Formally, a matched example of $\mathbf{x}_t$ is defined to be:

$$m_1(\mathbf{x}_t) = \arg \max_{\mathbf{x}_{t'} \in \mathbb{D}(T=t')} s(\phi(\mathbf{x}_t), \phi(\mathbf{x}_{t'})) \quad (5)$$

The *candidate set* is denoted by $\mathbb{D}(T = t')$ and is also known as the control group, contains only texts whose treatment variable equals t' , and is known in advance (i.e., both the treatment and control groups are pre-defined and there is no need to infer which text belong to which group). $\phi(\cdot)$ is a feature extractor function (the language representation model we aim to learn) that maps discrete text to the hypersphere space where matching is conducted by ranking candidates according their similarity with the query: $s(\phi(\mathbf{x}_t), \phi(\text{candidate}))$. The high-dimensional representation $\phi(\cdot)$ of each text is extracted by a text encoder and is the mean vector of the last layer hidden states of the text’s tokens. In our experiments, the similarity function $s(\cdot)$ is the cosine similarity:

$$s(\phi(\mathbf{x}_t), \phi(\mathbf{x}_{t'})) = \frac{\phi(\mathbf{x}_t)^T \phi(\mathbf{x}_{t'})}{\|\phi(\mathbf{x}_t)\| \cdot \|\phi(\mathbf{x}_{t'})\|} \quad (6)$$

Notice that Eq 5 defines *1:1 matching*, for *Top-K matching* we use: $\frac{1}{K} \sum_{i=1}^K \widehat{\text{ICaCE}}_f(\mathbf{x}_t, m_i(\mathbf{x}_t))$.

Our causal representation method is based on an Encoder-only language model (e.g., RoBERTa (Liu et al., 2019) or a SentenceTransformer (Reimers & Gurevych, 2019) in our case) fine-tuned to minimize an objective consisting of six contrastive loss components. The goal of the objective is to

enhance the similarity with approximate counterfactuals and matches while reducing similarity with misspecified counterfactuals and matches.

Given a causal graph, an intervention on a treatment variable $T : t \rightarrow t'$ and an example $\mathbf{x}_t$, we define the four following sets (see toy examples in Table 1):

1. $\mathbb{X}_{\text{CF}}$ - This set contains *Counterfactuals*, obtained by modifying $\mathbf{x}_t$ through the intervention on T , while the other variables are held unchanged. We utilize an LLM (ChatGPT) to simulate several CFs (up to ten).
2. $\mathbb{X}_{\text{M}}$ - This set contains *Matched samples*. These instances share identical values for the adjustment variables (that satisfy the back-door criterion, see Appendix §C). Our study employs a smaller model to predict their values and construct a set of valid matches.
3. $\mathbb{X}_{\text{-CF}}$ - This set contains *Misspecified Counterfactuals*. These examples are derived from editing $\mathbf{x}_t$ through interventions on variables other than the treatment variable. The term *misspecification* is well-known in the causal literature and indicates a wrong assumption of the DGP (Vansteelandt et al., 2012). In our case, a wrong intervention.
4. $\mathbb{X}_{\text{-M}}$ - This set contains *Misspecified Matched samples*. This is the complementary set to $\mathbb{X}_{\text{M}}$, containing all those instances that do not qualify as valid matches due to differing values of the adjustment variables.

We next formulate the *contrastive loss* Wang & Liu (2021), which aims to attract some input $\mathbf{x}$ to the “positive set” $\mathbb{X}_+$, and separate it from the “negative set” $\mathbb{X}_-$. τ is a temperature hyperparameter.

$$\mathcal{L}(\mathbf{x}, \mathbb{X}_+, \mathbb{X}_-) = -\log \left[\frac{\sum_{\mathbf{x}_+} \exp(s(\phi(\mathbf{x}), \phi(\mathbf{x}_+))/\tau)}{\sum_{\mathbf{x}_+} \exp(s(\phi(\mathbf{x}), \phi(\mathbf{x}_+))/\tau) + \sum_{\mathbf{x}_-} \exp(s(\phi(\mathbf{x}), \phi(\mathbf{x}_-))/\tau)} \right] \quad (7)$$

The final loss consists of six contrastive loss components:

$$\begin{aligned} \mathcal{L}(\mathbf{x}_t) = & \mathcal{L}(\mathbf{x}_t, \mathbb{X}_{\text{CF}}, \mathbb{X}_{\text{-M}}) + \mathcal{L}(\mathbf{x}_t, \mathbb{X}_{\text{CF}}, \mathbb{X}_{\text{-CF}}) + \mathcal{L}(\mathbf{x}_t, \mathbb{X}_{\text{CF}}, \mathbb{X}_{\text{M}}) \\ & + \mathcal{L}(\mathbf{x}_t, \mathbb{X}_{\text{M}}, \mathbb{X}_{\text{-M}}) + \mathcal{L}(\mathbf{x}_t, \mathbb{X}_{\text{M}}, \mathbb{X}_{\text{-CF}}) \\ & + \mathcal{L}(\mathbf{x}_t, \mathbb{X}_{\text{-CF}}, \mathbb{X}_{\text{-M}}) \end{aligned} \quad (8)$$

We hypothesize (and verify) that omitting components from the objective risks favoring inaccurate matches, as we highlight in the ablation study at §5.1. Accordingly, the objective prioritizes text matches that closely resemble CF approximations within a candidate set, defaulting to adjusting variable values in their absence. When choosing between misspecified CFs or misspecified matches, the methods favor the latter, given their potential shared traits with the CF, such as syntax or style linked to exogenous variables. Formally, given a query example $\mathbf{x}_t$ and four matching candidates $\mathbf{x}_{\text{CF}} \in \mathbb{X}_{\text{CF}}$, $\mathbf{x}_{\text{M}} \in \mathbb{X}_{\text{M}}$, $\mathbf{x}_{\text{-CF}} \in \mathbb{X}_{\text{-CF}}$, the candidates’ ranking, which is based on their similarity with $\mathbf{x}_t$, should be $\mathbf{x}_{\text{M}} \preceq \mathbf{x}_{\text{-CF}} \preceq \mathbf{x}_{\text{M}} \preceq \mathbf{x}_{\text{CF}}$. This desirable order is not arbitrary and is a direct product of the six components, ensuring robustness to variations of the candidate set. In addition, we investigate in the ablation study §5.1 the applicability of our matching method without pre-identified concept annotations in the training dataset (i.e., completely unsupervised setup). To this end, we utilize an LLM and predict the concept values in a zero-shot manner.

Training procedure. We start with a small set of textual examples (the train set) labeled with the adjusting variables (e.g., review concepts such as food and service). Initially, we employ a small Encoder-only model (RoBERTa) and train a concept predictor for each variable. The predictor models are then harnessed to construct the four sets: By predicting the concept values of each training example, we divide them (according to their concept values) and construct the $\mathbb{X}_{\text{M}}$ and $\mathbb{X}_{\text{-M}}$ sets.

For constructing $\mathbb{X}_{\text{CF}}$ and $\mathbb{X}_{\text{-CF}}$, we use a few-shot LLM (ChatGPT) to generate approximate CFs and misspecified CFs (see prompts at §D.1). We filter out misspecified CFs from $\mathbb{X}_{\text{CF}}$ if the concept predictors indicate an adjusted variable was also changed during the treatment intervention. Additionally, we use simple rules to filter unsuccessful generations (e.g., empty strings, “As an AI LM...”).

To train our language representation model (aimed to explain a specific concept), we proceed as follows: For every training example $\mathbf{x}_t$, we randomly sample four examples: $\mathbf{x}_{\text{CF}} \in \mathbb{X}_{\text{CF}}$, $\mathbf{x}_{\text{M}} \in \mathbb{X}_{\text{M}}$, $\mathbf{x}_{\text{-CF}} \in \mathbb{X}_{\text{-CF}}$. The training goal is to minimize the objective of Eq. 8. We repeat this process for several epochs (15) and select the checkpoint that achieves the lowest loss on the dev. set.

4 EXPERIMENTAL SETUP

4.1 EXPLANATION METHOD EVALUATION

Causal Estimation-Based Benchmark (CEBaB). CEBaB (Abraham et al., 2022) comprises high-quality, labeled CFs designed to benchmark model explanation methods. This benchmark originated from thousands of original restaurant reviews obtained from OpenTable. For every original review, human annotators were tasked to edit the review and write an approximate CF that reflects a specific intervention, e.g., “change the service concept from positive to negative”. Each entry (original reviews and approximate CFs) in CEBaB received a 5-star sentiment rating, labeled by five annotators. Furthermore, every text was annotated at a conceptual level, as positive, negative or neutral w.r.t. four central mediating concepts: Food, Service, Ambiance, and Noise (see Figure 1). CEBaB contains two train sets, exclusive ($N = 1463$) and inclusive (which we do not use), development ($N = 1672$), and test ($N = 1688$) sets. We randomly split the exclusive set into two equal sets: the train set (for training the causal representation model and the concept predictors) and a matching candidates set.

Evaluation pipeline. A high-level explanation method that provides an estimator of the ICaCE can be evaluated using benchmarks like CEBaB, i.e., an interventional dataset consisting of examples, interventions, and a corresponding human-written ground-truth CF for each example and intervention. For a given example $\mathbf{x}_t$ and an intervention $T : t \rightarrow t'$, using the example the corresponding ground-truth CF $\tilde{\mathbf{x}}_{t'}^h$ we can estimate the golden individual causal effect: $y^h = \widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^h)$

For the same example $\mathbf{x}_t$ and intervention, the explanation method estimates the individual causal effect y^m . In the case of a CF-based explanation method, the estimation is computed with the approximated CF: $y^m = \widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^m)$. For the generative approach, we generate a CF $\tilde{\mathbf{x}}_{t'}^m$ using an LLM and calculate $\widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^m)$. For matching, given an intervention $T \leftarrow t'$ and a set of candidates that share the new treatment value: $\mathbb{D}(T = t') = \{\mathbf{x} | \mathbf{x} \in \mathbb{D}, T(\mathbf{x}) = t'\}$, we utilize a matching model (e.g., our causal representation model) to represent each candidate with a high-dimensional continuous vector. Then, given a query example $\mathbf{x}_t$, we extract its representation, rank the candidates according to their cosine similarity, select a match, and calculate the $\widehat{\text{ICaCE}}_f(\mathbf{x}_t, m_1(\mathbf{x}_t))$.

Evaluation measures. Given the two estimations, y^h and y^m , we then calculate the distance between the two using three metrics introduced in Abraham et al. (2022) $L2$ distance, Cosine distance, and norm difference.

$$L2(y^h, y^m) = \|y^h - y^m\|_2 \quad (9)$$

$$\text{Cos}(y^h, y^m) = 1 - \frac{(y^h)^T y^m}{\|y^h\|_2 \|y^m\|_2} \quad (10)$$

$$\text{ND}(y^h, y^m) = \left| \|y^h\|_2 - \|y^m\|_2 \right| \quad (11)$$

Finally, the distance is plugged into Eq.12 to provide the estimation error of the explanation method:

$$\text{Err}(f, m, T, t, t') = \frac{1}{|\mathbb{D}(T \leftarrow t')|} \sum_{(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^h) \in \mathbb{D}} \text{Dist} \left(\widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^h), \widehat{\text{ICaCE}}_f(\mathbf{x}_t, \tilde{\mathbf{x}}_{t'}^m) \right) \quad (12)$$

The Err scores we present in this study are the mean over 24 interventions: four concepts with three values (six interventions for each concept).

4.2 MODELS AND BASELINES

In Appendix D we provide additional implementation details, including hyperparameters and prompts.

Generative approach. We employ three different CF generation techniques: *Zero-shot Generative*, *Few-shot Generative* (with three demonstrations) and *Fine-tune Generative*. The first two are based on the Decoder-only ChatGPT (Ouyang et al., 2022), while the third is based on an Encoder-decoder T5-base model (Raffel et al., 2019) trained using a parallel dataset, comprised of original reviews and their human-written CFs (the inclusive train set of CEBaB). The zero-shot and few-shot prompts are described in Appendix §D.1.

Matching baselines. We benchmark against six matching methods: (1) *Random Match*, which randomly selects an example from the candidate set; (2) *Propensity* (Benedetto et al., 2018), which

Table 2: Comparison between different methods and baselines. The columns present Err scores (Eq. 12) when explaining different five-class sentiment models. The sub-columns present three different measures: Euclidean distance (*L2*), Cosine distance (*Cos*), and norm difference (*ND*). The top table presents scores using a single match ($K = 1$), and the bottom table when $K = 10$. The *Generative* rows are not matching methods and thus are not comparable to such methods. Another non-comparable row is the first, which presents the performance of our causal model with a candidate set that also includes ground-truth (GT) CFs. **Numbers are means over 24 interventions, ↓ is better.**

$K = 1$	DistilBERT			BERT			RoBERTa			Llama2-7b			Llama2-13b			AVG
	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	
Causal Model (+GT)	.13	.09	.06	.14	.10	.07	.13	.09	.06	.12	.07	.06	.11	.07	.06	.09
Fine-tune Generative	.38	.28	.21	.42	.28	.23	.39	.27	.21	.36	.23	.22	.31	.21	.19	.28
Few-shot Generative	.42	.32	.23	.45	.29	.23	.41	.30	.21	.38	.26	.23	.33	.23	.20	.30
Zero-shot Generative	.43	.35	.24	.47	.35	.25	.44	.34	.23	.40	.27	.25	.35	.26	.21	.32
Random Match	.88	.64	.47	.96	.62	.49	.95	.63	.49	.84	.54	.50	.84	.55	.49	.66
Propensity	.90	.68	.48	.99	.69	.56	.99	.68	.56	.88	.61	.53	.86	.58	.52	.70
Approx	.74	.60	.41	.81	.58	.45	.79	.60	.44	.71	.49	.43	.70	.54	.49	.58
PT RoBERTa	.75	.61	.39	.83	.59	.42	.81	.61	.41	.74	.53	.43	.73	.51	.43	.59
PT S-Transformer	.75	.60	.40	.83	.60	.45	.81	.61	.43	.74	.55	.44	.69	.52	.41	.59
FT S-Transformer	.72	.78	.46	.82	.79	.59	.80	.84	.56	.73	.63	.45	.68	.63	.39	.66
Causal Model	.66	.55	.36	.70	.55	.39	.68	.56	.37	.64	.47	.39	.59	.45	.36	.52
$K = 10$	DistilBERT			BERT			RoBERTa			Llama2-7b			Llama2-13b			AVG
	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	<i>L2</i>	<i>Cos</i>	<i>ND</i>	
Fine-tune Generative	.32	.21	.20	.36	.23	.23	.34	.21	.21	.30	.18	.20	.27	.16	.18	.24
Few-shot Generative	.38	.30	.22	.42	.28	.24	.38	.29	.21	.34	.24	.22	.30	.22	.19	.28
Zero-shot Generative	.38	.32	.23	.42	.31	.25	.40	.31	.23	.36	.24	.23	.31	.22	.19	.29
Random Match	.70	.52	.40	.78	.51	.47	.77	.51	.45	.67	.45	.40	.67	.45	.40	.54
Propensity	.72	.54	.43	.80	.52	.51	.79	.52	.50	.68	.47	.42	.69	.47	.43	.57
Approx	.61	.49	.37	.68	.48	.43	.66	.48	.41	.58	.40	.37	.57	.41	.36	.49
PT RoBERTa	.64	.50	.36	.71	.49	.44	.70	.49	.42	.61	.44	.37	.60	.43	.36	.51
PT S-Transformer	.64	.51	.38	.72	.50	.45	.71	.50	.44	.61	.45	.38	.59	.44	.36	.51
FT S-Transformer	.61	.56	.43	.69	.56	.50	.67	.56	.47	.60	.48	.41	.56	.50	.36	.53
Causal Model	.56	.47	.35	.63	.46	.41	.60	.45	.38	.55	.40	.36	.52	.38	.33	.46

first estimates the propensity score $P(T = t'|x)$ using a concept predictor, and then conducts matching based on this score; (3) *Approx*, which randomly selects an example from $\mathbb{X}_M$. Notice that $\mathbb{X}_M$ can be an empty set, thus, unlike the other baselines, matching may not be performed; (4) *PT RoBERTa*, which utilizes a pre-trained LM to represent texts and finds matches using their cos similarity; (5) *PT S-Transformer*, same as PT RoBERTa, expect the backbone model is a SentenceTransformer model (Reimers & Gurevych, 2019) trained to maximize semantic textual similarity; and (6) *FT S-Transformer*, a SentenceTransformer fine-tuned to predict the five-star sentiment; Notice that Abraham et al. (2022) found that Approx outperforms all seven tested (non-matching) baselines (see Figure 6). Therefore we do not include them in this study.

Interpreted models. We interpret five different models of varying sizes. The first three are Encoder-only models fine-tuned using only the original reviews from the train set of CEBaB to predict the five-star sentiment. These models include DistilBERT (Sanh et al., 2019), BERT (Devlin et al., 2019), and RoBERTa (Liu et al., 2019). The other two are the zero-shot chat versions of the Decoder-only Llama-2 model (Touvron et al., 2023) with 7 and 13 billion parameters. We extract the five-star sentiment distribution of the Llama models using the prompt described in Figure 7.

5 RESULTS

Our main results are provided in Table 2. In Appendix §5.1 we provide a thorough ablation study, demonstrating the effect of each of our design choices, including, among others, the impact of variations of the candidate set and the causal model objective (Eq. 8). Our three main findings are:

1. Counterfactual generation models are the SOTA model-agnostic explainers. The first promise of this study is that CF generation models are strong explainers. As can be seen from Table 2, the generative models achieve much lower errors compared to the other methods, making them

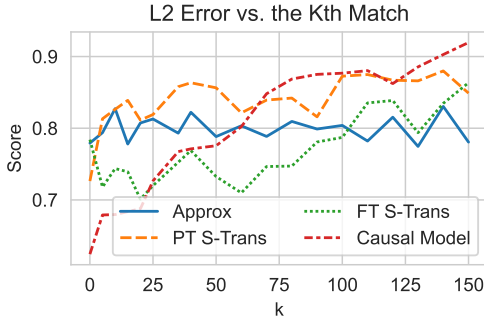


Figure 2: The L2 Err score for explaining the DistilBERT model (Y-axis) as a function of selecting the k^{th} match, $m_k(x)$. For Figures 2 and 3, we present average Err scores only for interventions with a candidate set of size larger than 250, making the numbers differ from Table 2.

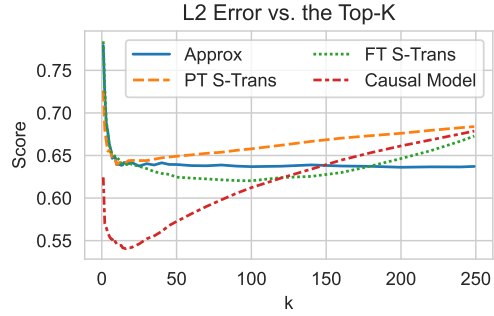


Figure 3: The L2 Err score for explaining the DistilBERT model (Y-axis) as a function of Top- K matching. The figure illustrates the beneficial impact of considering multiple matches. However, beyond a certain point, adding more matches negatively affects causal estimation.

the SOTA model-agnostic explainers. In addition, the table provides two additional findings: (1) Few-shot prompts are better than zero-shot prompts for generating CFs; and (2) The small fine-tuned Encoder-decoder T5 is the best generative model. This suggests that when a parallel dataset consisting of pairs of input examples and their human-written CFs is available, then it is better to fine-tune a small model. Moreover, the small model is computationally (and financially) less expensive than LLMs (but much slower than matching methods). Nevertheless, consider that collecting a parallel dataset for fine-tuning is labor intensive.

2. The causal representation model is the best-performing matching method. Table 2 also sheds light on the performance of our novel causal representation method (described in §3.4, *our causal model* in short) in comparison to various explainers and matching baselines. Notably, our causal model consistently outperforms all the matching baselines, achieving substantially lower errors across five explained models. Noteworthy, Abraham et al. (2022) proposed the Approx method (the only matching method they examined) and found it outperforms common NLP explainers. Our results show that all matching baselines (including Approx, excluding Random Match and Propensity) are competitive and perform similarly. Nevertheless, they fall short of our causal representation model.

Although the generative approach exhibits better scores than the matching approach, it is important to note that the quality of the $\widehat{\text{ICaCE}}_f$ estimation of any matching method is highly dependent on the matching set. Therefore, we also include the Err scores for our causal model when the candidate set contains the ground-truth human-written CFs (see the first row in Table 2). In that case, our causal model outperforms even the generative models. This suggests that our method is potentially the SOTA explainer (although this is impractical since access to CFs is a strong assumption).

Finally, Figure 2, which plots the Err score when selecting the k^{th} match, $m_k(x)$, reveals three desirable properties of our causal model: (1) The lowest Err score belongs to the first match, indicating that, on average, our causal model indeed selects the best match; (2) A monotonically increasing pattern (i.e., a larger k results in a higher Err) demonstrates a strong association between representation similarity and explainability capacity: The more similar a match is, the more accurate the $\widehat{\text{ICaCE}}_f$ estimator becomes; and (3) The top-ranked matches selected by our causal model outperform other baselines, achieving notably lower Err scores.

3. Top- K matching improves the explanation capabilities of every method. When comparing the top rows of Table 2 ($k = 1$) to the bottom rows ($k = 10$), it is easy to observe that Top- K matching lowers the Err of every examined explanation method. This observation also holds for the generative models, which accordingly generate multiple CFs via sampling. The definitive results, combined with the simplicity and cost-effectiveness of the Top- K extension (especially for matching methods, see Appendix §B), suggest that top- K should be a standard for explainers. Given this, the question that arises is *what should the value of K be?*

Table 3: Ablation study. Row 1: Our model. Row 2: With a different backbone. Row 3: Causal S-Transformer (ours) without filtering misspecified CFs from $\mathbb{X}_{CF}$. Row 4: Without human-annotated concepts (i.e., concepts are predicted by a zero-shot LLM). Rows 5-10: Different variations of the objective (Eq. 8) when discarding components that include the specified sets. Columns present different matching candidate sets: original, original with ground-truth CFs, and original with misspecified CFs. **Numbers are the mean over 5 explained models and 24 interventions.** $\downarrow$ is better.

		Original			+ GT CFs			+ Miss. CFs		
		<i>L2</i>	<i>Cos</i>	<i>norm</i>	<i>L2</i>	<i>Cos</i>	<i>norm</i>	<i>L2</i>	<i>Cos</i>	<i>norm</i>
1	Causal S-Trans.	.65	.52	.38	.13	.09	.06	.70	.58	.46
2	Causal Roberta	.66	.52	.37	.16	.11	.08	.70	.59	.46
3	w/o filtering	.66	.51	.38	.12	.08	.05	.77	.63	.52
4	w/o labels	.65	.52	.36	.13	.09	.06	.67	.59	.43
5	w/o $\mathbb{X}_{CF}$	.67	.54	.39	.26	.18	.13	.69	.58	.45
6	w/o $\mathbb{X}_M$	.66	.51	.37	.09	.06	.04	.81	.68	.57
7	w/o $\mathbb{X}_{-CF}$	.66	.51	.38	.06	.04	.03	.83	.68	.60
8	w/o $\mathbb{X}_{-M}$	.66	.51	.38	.16	.11	.07	.70	.59	.46
9	w/o $\mathbb{X}_M \cup \mathbb{X}_{-M}$	.66	.53	.39	.19	.13	.09	.74	.61	.50
10	w/o $\mathbb{X}_{CF} \cup \mathbb{X}_{-CF}$	.66	.52	.38	.15	.09	.07	.72	.60	.49

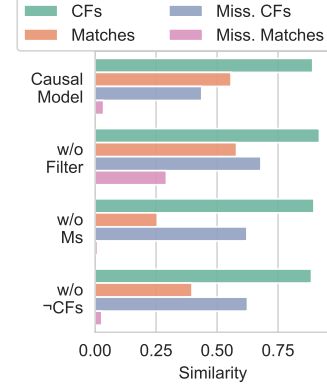


Figure 4: Average similarity between query examples and examples from the four sets (the bars), plotted for different variants of the causal model (Y-axis).

In Figure 3, we plot the L2 Err as a function of k for our method and three other baselines when explaining DistilBERT (the trends are similar for the other metrics and explained models). The prominent ‘✓’ shape curve of the causal model is desirable and indicates that the model selects good matches at first. Then, around $k = 20$, the error begins to surge and only after $k = 100$ that it matches the error at $k = 1$. Conversely, other baselines rank candidates less effectively and exhibit a more gradual increase after their minimum. Nonetheless, the Top- K approach has a beneficial effect for all methods and a broad range of values can be picked before the effect fades. We arbitrarily opted for $k = 10$ in this study, even though $k = 20$ would have been more advantageous for our causal model. A future direction could explore how to tune the k parameter for each individual example.

5.1 ABLATION STUDY

This subsection aims to shed more light on the effect of each of our design choices. To this end, we examine variations or omissions of components from our causal representation model. Noteworthy, with the original candidate set, which is relatively small, all of the ablation models are competitive, and the performance difference is insignificant. Therefore, we also experiment with variations of the candidate set that reveal the pitfalls of each ablation model. Table 3 presents the performance when using three candidate sets: (1) The original candidate set, containing only original reviews from the matching split; (2) The original set augmented by the ground-truth human-written CFs; and (3) The original set augmented by the misspecified model-generated CFs.

We start by examining the *backbone encoder model* of the causal model: SentenceTransformer and RoBERTa. Comparing the first two rows of Table 3, we observe that both encoders are competitive and can be utilized as the backbone. We decided to use SentenceTransformer because its development loss was lower. By comparing the first row to the third, we examine the impact *filtering unsuccessful or misspecified CFs* from $\mathbb{X}_{CF}$. Without filtering, the causal model prioritizes misspecified CFs above valid matches (see Figure 4), and as a result, when misspecified CFs are present in the candidate set, the performance degrades (third row, right column, gray cells).

In row 4, we investigate the applicability of our method without human-annotated concepts in the training dataset (i.e., completely unsupervised setup). Utilizing an LLM (ChatGPT), we predicted concept values in a zero-shot manner. Row 4 reveals that the performance remains consistent and on par with models trained on human-annotated datasets. Surprisingly, it performs better in some metrics (although insignificant), likely because the LLM predicts annotations that are sometimes missed due to disagreements between annotators.

Table 4: Examples from the new Stance Detection setup. In the first stage of the dataset construction (top table), we split the original tweets into the train, dev, matching, and test sets. In the second stage (middle table), we keep 30% of the original tweets, and for the remaining 70%, we randomly assign values for the three writer’s concepts. In this example, we ask GPT-4 to edit the tweet “to make it sound as if a teenage female wrote it”. In the third (bottom table), we generate CFs for the test set (which are used for benchmarking). Here, we ask GPT-4 to change the Job concept to Farmer.

S1: Original tweet. I need feminism because “what were you wearing” shouldn’t be a question when I tell my stepmom I was cat called. #SemST				
<i>Subject</i> : Feminism	<i>Age</i> : ?	<i>Gender</i> : ?	<i>Job</i> : ?	<i>Label</i> : Favor
S2: Modified tweet. OMG, like, I totally need feminism cuz ‘what were you wearing?’ shouldn’t even be a thing when I tell my stepmom some guy shouted stuff at me. #JustSaying				
<i>Subject</i> : Feminism	<i>Age</i> ← Teen	<i>Gender</i> ← ♀	<i>Job</i> : ?	<i>Label</i> : Favor
S3: CF (test set). Gosh, I reckon I’d holler for feminism, cause ‘what were ya wearin?’ ought not be a question when I tell ma about some fella yelled stuff at me while I was out in the fields. #JustSaying				
<i>Subject</i> : Feminism	<i>Age</i> : Teen	<i>Gender</i> : ♀	<i>Job</i> ← Farmer	<i>Label</i> : Favor

We next examine the impact of *discarding components from the model objective Eq 8*. To this end, for each set of examples, $\mathbb{X}_{CF}$, $\mathbb{X}_M$, $\mathbb{X}_{-CF}$, and $\mathbb{X}_{-M}$, we train an encoder model while ignoring the three components from the objective that utilize examples from the discarded set. In addition, we also train two encoder models using only model-generated CFs (w/o $\mathbb{X}_M$ and $\mathbb{X}_{-M}$), and without any (w/o $\mathbb{X}_{CF}$ and $\mathbb{X}_{-CF}$, i.e., training only with human-written reviews – not CFs).

The behavior of each ablation model is as one would expect. For example, when $\mathbb{X}_{CF}$ is discarded from the objective (row 4), the ablation model struggles to identify CFs and the performance declines when the matching sets include ground-truth CFs. Conversely, when $\mathbb{X}_{-CF}$ or $\mathbb{X}_M$ are excluded (rows 6, 7), the model fails to distinguish between CFs and misspecified ones, often favoring the latter over valid matches. Consequently, the performance drops when the candidate set contains misspecified CFs. *Only our causal model consistently performs well across both candidate sets*. Notably, we can see that even without model-generated CFs (row 9), the causal model performs well. Although the causal estimation is less precise, *achieving a good explainer without utilizing an LLM is possible*.

Finally, Figure 4 presents the average similarity between query examples and the four sets for the causal and selected ablative models. As can be seen, our causal model learns the desirable ranking order: $x_{-M} \preceq x_{-CF} \preceq x_M \preceq x_{CF}$. In contrast, the model learns to favor misspecified CFs over matches when discarding $\mathbb{X}_M$ or $\mathbb{X}_{-CF}$ or when not performing filtering of misspecified CFs from $\mathbb{X}_{CF}$. Although these ablation models have a competitive performance to our model and some outperform it when the candidate set contains ground-truth CFs, they fail when the set contains misspecified CFs, highlighting the importance of learning the right ranking order.

6 NEW SETUP: EXPLAINING STANCE DETECTION MODELS

While we aim to validate our earlier conclusions across multiple benchmarks, CEBaB remains the sole high-quality benchmark for NLP explainers. On the other hand, we show that LLMs generate CFs that resemble human-written CFs. Consequently, we harness LLM capabilities to emulate CEBaB and propose a new benchmark for model explanation, eliminating the high cost of manually annotating such benchmarks. We decided to focus on Stance Detection, which is an important and well-studied NLP task that determines the position (support, oppose, neutral) expressed by the writer towards a particular topic (e.g., abortions, climate change) (Küçük & Can, 2021; Hardalov et al., 2022; Viehmann et al., 2023). This task is intriguing because understanding public opinions and beliefs can guide decision-making, policy formulation, and marketing strategies. Estimating the causal effect of high-level concepts like gender or age may shed light on the societal biases and perceptions that drive social trends.

We rely on the stance detection dataset from the SemEval 2016 shared task (Mohammad et al., 2016). This dataset contains tweets about five subjects: Abortion, atheism, climate change, feminism, and Hillary Clinton. Humans annotated the stance of the tweets with three possible labels: support, oppose, and neutral. For simplicity of the new setup, we embrace the same causal graph of CEBaB

Table 5: Results for the new stance detection setup we introduce in an out-of-distribution scenario (for each test example, the candidate set compromise texts with different subjects). We explain two stance detection models: **New** - RoBERTa fine-tuned on new stance labels extracted by GPT-4; and **Original** - DistilBERT fine-tuned with the stance labels of the original tweets. The description of the rows is given in the caption of Table 2. **Numbers are the mean over 18 interventions, ↓ is better.**

	$K = 1$						$K = 10$						
	New Labels			Original			New Labels			Original			AVG
	$\underline{L2}$	$\underline{Cos}$	$\underline{ND}$	$\underline{L2}$	$\underline{Cos}$	$\underline{ND}$	$\underline{L2}$	$\underline{Cos}$	$\underline{ND}$	$\underline{L2}$	$\underline{Cos}$	$\underline{ND}$	
Causal Model (+GT)	.08	.16	.06	.10	.13	.08	-	-	-	-	-	-	0.10
Zero-shot Generative	.22	.74	.16	.36	.65	.30	.19	.68	.15	.34	.61	.30	.39
Random Match	.57	.85	.45	.85	.77	.71	.46	.82	.36	.76	.73	.65	.66
Propensity	.55	.87	.44	.83	.80	.70	.44	.83	.34	.74	.74	.64	.66
Approx	.54	.87	.43	.73	.75	.60	.43	.82	.34	.65	.72	.56	.66
PT RoBERTa	.47	.89	.36	.76	.75	.63	.41	.84	.31	.70	.73	.60	.62
PT S-Transformer	.47	.86	.36	.77	.75	.64	.40	.81	.30	.73	.73	.63	.62
Causal Model	.46	.84	.36	.71	.74	.60	.40	.80	.30	.64	.71	.58	.59

(see Figure 1), but replace the four review concepts with new relevant concepts: (1) Tweet’s subject (specified above); (2) Writer’s age - teenage, elder or unknown; (3) Writer’s gender - female, male or unknown; and (4) Writer’s Job - farmer, professor or unknown. The four concepts have an exogenous variable U as their parent and are the direct cause of the text X and the label (stance) Y , which is naturally affected by the four concepts. The explanation methods should estimate the effect of 9 interventions: changing the value of the three writer’s concepts (age, gender, job) to three values (we do not explain the tweet’s subject concept).

Constructing the new setup. We follow three steps for constructing the new benchmark. See the examples in Table 4. In the first stage, we randomly split the data into four sets: train ($N = 1000$), matching ($N = 2250$), dev ($N = 250$), and test ($N = 500$). Each subject is precisely one-fifth of each split. For each set and subject, we keep 30% of the original tweet and assign an ‘unknown’ value to the writer’s concepts. For the remaining 70%, we randomly sample two of the three writer’s concepts and assign new values (e.g., male, unknown age, professor). Noteworthy, we do not uniformly sample the writer’s profile since we want to introduce correlations between the concepts and the label. Table 7 presents the joint probabilities of different concept and label values.

In the second stage, we utilize GPT-4 to generate new tweets according to the new value assignment of 70% of the tweets. The prompts we used are described in Figure 10. In the third and final stage, we utilize GPT-4 to generate the ground-truth CFs for the test set, which enables calculating the Err for evaluating explanation methods. Accordingly, for each test example, we prompt GPT-4 and generate a single CF for each of the six possible interventions (3 writer’s concepts with two possible values that are different from the assignment). The prompt is provided in Figure 11. In addition, we ask GPT-4 to predict the new label of the tweet (after the edit) and add this information to the dataset.

Experimental details. We interpret two stance detection models. The first one is a RoBERTa fine-tuned on the new stance labels extracted by GPT-4. The second model is DistilBERT fine-tuned on the stance labels of the original tweets. Noteworthy, the causal effect of the writer’s concepts is more pronounced in the second model. This arises because we designed the dataset to have a large correlation between the original labels and the concepts, which the fine-tuned model captures. In contrast, when fine-tuning with the new labels produced by GPT-4, this causal effect becomes weak. This is mainly because GPT-4 tends to predict the neutral label far more frequently than its representation within the original labels (64.8% vs. 25.5%, refer to Table 7), thereby diminishing the correlation we initially established.

In contrast to CEBaB, our new setup enables testing of our causal matching method in an out-of-distribution scenario. A prevalent approach in the stance detection literature is assessing model robustness against distribution shifts arising from changes in the subject of the texts. For instance, a model might be trained on texts discussing abortion and then tested on texts about climate change (Herdalov et al., 2021; Ben-David et al., 2022; Deng et al., 2022). In our context, such a distribution shift might happen when the subject of the test samples is distinct from the subjects within the matching candidate set. Consequently, we conduct experiments to gauge the performance of our

method under circumstances where the distribution shift is evident (i.e., for each test example, the matching set compromises only candidates with different subjects).

For the CF generation approach, we employ ChatGPT (and not GPT-4, which generates the ground-truth CFs of the test set) and only with zero-shot prompts. We benchmark our causal representation model with the same matching baselines from §4.2 with the same hyperparameters and training procedure described in §D.

Results. We report the results on Table 5. As can be seen, we reproduce our main findings from the previous section on the new benchmark, confirming the robustness of our conclusions, even in out-of-distribution settings. Specifically, the generative approach outperforms the matching methods (although consider that in this setup, the ground-truth CFs are also model-generated). In addition, the causal representation model surpasses the other matching baseline and outperforms the generative model when the candidate set contains ground-truth CFs. Finally, the Top- K technique universally improves all the explainers.

7 REQUIREMENTS AND LIMITATIONS

In this section, we discuss the setup requirements for estimating the causal effect according to our two approaches and the limitations these requirements may pose. In addition, we discuss and demonstrate the applicability of our approaches to more complex causal graphs and setups than CEBaB.

7.1 REQUIREMENTS

Causal graph. The first requirement for any causal estimation method is access to the causal graph describing the concepts and the relationships between them. Since the causal graph is typically derived by a domain expert, specifying it might seem like a limitation preventing generalization and automation of our explanation methods. However, notice that causal graph specification is a requirement for accurate causal estimation (Shpitser & Pearl, 2006; Feder et al., 2021b;a), this also aligns with our theoretical result: Explanation methods that are unaware of the causal mechanism might be unfaithful. In some fields (such as healthcare), causal explanations are crucial, making the effort to construct a causal graph a worthwhile endeavour. Consider an NLP model that recommends medical treatments based on symptoms described in the medical record. In that case, the clinician can not rely on correlational interpretations of the model recommendations. Finally, following future research and technological advances, it is reasonable to assume that the reliance on human experts for concept and causal graph discovery will reduce, and this process will gradually become more and more automatic.

Annotated training dataset In the generative approach, a training dataset is not essential, except when performing few-shot prompting. In this case, a small selection of text examples, interventions, and counterfactuals is required.

Conversely, the general requirement of any matching method is the availability of a candidate set from which the method selects a match. The candidate set should contain an annotation of the treatment concept. This is because the method needs to select a match annotated with the corresponding target intervention value; otherwise, it will fail to provide an accurate estimation. In our setup, the candidate set is randomly sampled from the training set. For training our causal representation method, we also need a training set annotated with the adjusted variables. This set is used for: (1) Constructing the $\mathbb{X}_{CF}$ and $\mathbb{X}_{-CF}$ sets by prompting an LLM to generate CFs of examples from the training set; (2) Training concept predictors - which are used for filtering misspecified CFs; (3) Constructing the $\mathbb{X}_M$ and $\mathbb{X}_{-M}$ sets.

Although a candidate set and a training dataset with annotations are used in our experiments, we believe they are not required and can be easily generated by an LLM. We demonstrate this approach in our ablation study (subsection §5.1), where we train our matching method with concept annotations predicted by an LLM. According to the results, the performance remains consistent and on par with models trained on human annotations.

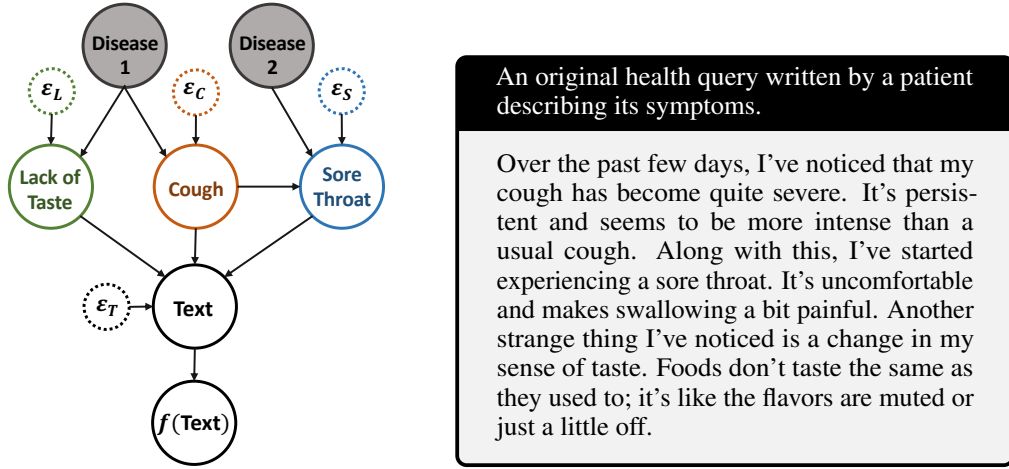


Figure 5: The causal graph of our motivating case study of a patient-doctor interaction in an online health consultation. A text-based health query (*Text*, see example on the right) is analyzed by a doctor with the assistance of an NLP classifier that outputs a diagnosis of possible diseases. In this causal graph, three symptoms affect the patient’s text, with a notable interrelationship between *Cough* and *SoreThroat*. The exogenous variables impacting the symptoms are $\varepsilon_L, \varepsilon_C, \varepsilon_S, \varepsilon_T$.

7.2 COMPLEX CAUSAL GRAPHS

Our choice of CEBaB was driven by its unique status as the only non-synthetic interventional dataset tailored for benchmarking concept-level explanation methods in NLP. In this subsection, we discuss and elaborate on the applicability of our theorem and methods to other causal graphs beyond CEBaB.

First, our theorem is designed to be flexible and does not rely on a specific type of causal graph. Our proof is based on a minor modification to the graph, adding a confounder, which can be applied to graphs of any complexity. The key observation is that while non-causal methods may struggle with this change, causal methods remain faithful. Thus, our theorem holds regardless of the complexity of the causal graph as long the causal effect can be identified from the graph. Second, our methods are also applicable to any identifiable causal graph. The identification requirement is the only limitation of applying our methods. However, without further assumptions on the causal mechanism, it is also a limitation of any causal estimation method. For an example of a causal graph that is non-identifiable see Figure 4 in Cinelli et al. (2022).

We next provide proof of concepts that strong LLMs (such as GPT-4) can succeed in generating CFs for graphs that differ from CEBaB, especially those involving mediators. By incorporating a mediator into the causal graph, we aim to explore situations where an intervention leads to changes in more than just the outcome text. Our case study is based on the causal graph illustrated in Figure 5 and involves an online health consultation between a patient and a doctor. The patient submits a written health query, and the doctor then reviews this query, aided by an NLP classifier that estimates the likelihood of various diseases. Based on an explanation method that quantifies the effect of each symptom on the prediction of the disease, the doctor can continue her interaction with the patient and concentrate on specific symptoms for a more accurate diagnosis.

According to the causal graph in Figure 5, there are three symptoms: *Lack of Taste*, *Cough*, and *Sore Throat*. While *Disease 1* causes a Lack of Taste and Cough, *Disease 2* causes a Sore Throat. However, any combination of disease and symptoms is possible due to the involvement of noise originating from the exogenous variables. In addition, an intensive cough can also cause a sore throat. Hence, when Cough is the concept we explain, Lack of Taste is a confounder, while Sore Throat is a mediator. The original query of the patient is provided on the right side of Figure 5, describing a patient suffering from all three symptoms.

In the examples below, we show how we can utilize GPT-4 to generate CFs by intervening on the cough symptom to estimate the direct causal effect (red box) or the total causal effect (green box).

GPT-4 prompt for direct effect estimation: Generate a counterfactual by making the patient’s cough weaker. Keep the symptom of lack of taste and sore throat fixed.

Ex1: Over the past few days, I’ve noticed a **mild cough. It’s not very persistent and seems less intense than a usual cough.** Along with this, I’ve started experiencing a sore throat. It’s uncomfortable and makes swallowing a bit painful. Another strange thing I’ve noticed is a change in my sense of taste. Foods don’t taste the same as they used to; it’s like the flavors are muted or just a little off.

Ex2: Over the past few days, I’ve noticed a **mild cough. It’s intermittent and not as intense as a usual cough.** Along with this, I’ve started experiencing a sore throat. It’s uncomfortable and makes swallowing a bit painful. Another strange thing I’ve noticed is a change in my sense of taste. Foods don’t taste the same as they used to; it’s like the flavors are muted or just a little off.

GPT-4 prompt for total effect estimation: Generate a counterfactual by making the patient’s cough weaker. Keep the symptom of lack of taste fixed. Notice that a cough could impact the sore throat.

Ex1: Over the past few days, I’ve noticed a **mild cough. It’s not very persistent and seems less intense than a usual cough. Despite the mildness of the cough, I’ve started experiencing a sore throat.** It’s uncomfortable and makes swallowing a bit painful, though not severely so. What’s more peculiar is a change in my sense of taste. Foods don’t taste the same as they used to; it’s like the flavors are muted or just a little off.

Ex2: Over the past few days, I’ve noticed that **my cough is present but not particularly severe. It’s more like a mild, occasional cough rather than a persistent or intense one. Along with this, there’s a slight soreness in my throat, but it’s not too uncomfortable and doesn’t cause much pain when swallowing.** Another strange thing I’ve noticed is a change in my sense of taste. Foods don’t taste the same as they used to; it’s like the flavors are muted or just a little off.

To generate approximate CFs for estimating the direct causal effect (red box), we adjust for the Lack of Taste and the Sore Throat concepts since we are only interested in the direct path between Cough and the text. We do this by prompting GPT-4 to keep these variables fixed. The red box shows GPT-4 successfully generates CFs by modifying the text describing the cough symptom while leaving the remaining text fixed.

In the case of total effect estimation (green box), we should not adjust for the mediator concept - Sore Throat. This is because a change in Cough may or may not cause a change in Sore Throat (which also affects the text). We can achieve this by informing GPT-4 that a change in cough could also impact sore throat. As the two CFs in the green box show, GPT-4 sometimes ignores a potential change in the sore throat symptom (Ex1) and sometimes modifies the relevant text (Ex2). This demonstrates that strong LLMs like GPT-4 can handle complex situations when an intervention not only directly affects the text but also may impact other concepts.

Notice, however, that relying on a single CF might lead to a biased causal effect estimation. Therefore, generating multiple CFs is crucial. This perfectly aligns with our theoretical definition of an approximated CF explanation method which considers an approximation error. Accordingly, utilizing multiple approximated CF for estimating the causal effect may lower the variance of the estimator and make it more robust. This also explains why this technique (of Top-K matching) universally improves the performance of any CF-based explanation method in our study. Notice that in the case of mediators, LLMs may not model accurately the conditional distribution of the mediator given the intervention (e.g., $P(\text{SoreThroat}|\text{Cough})$ in our causal graph). We leave this challenge to future research, although a solution might be reweighing the CFs according to this probability.

Finally, our work underscores a novel contribution in utilizing LLMs for generating CFs. As we demonstrate, this can facilitate the creation of new interventional datasets representing complex causal graphs and significantly advance the research in causal explanations and benchmark construction.

8 DISCUSSION

In this study, we explored the terrain of explaining the impact of high-level concepts on the predictions of NLP models. We focused on model-agnostic techniques, which do not require access to the explained model during the training time and hence can explain numerous models, a property that is crucial for model selection, debugging, and deployment of safer, transparent, and accountable systems. We provided a theoretical framework linking faithfulness to causality, showing that CF approximation explanation methods are always order-faithful. We further motivate the utilization of LLMs for generating CFs, either as part of the explanation or for learning an embedding space that enables effective matching. We hope our theoretical and empirical results might broadly impact the field, and suggest these four following directions:

Explanation methods should be causal-inspired. Throughout this paper, our discussions as well as the theoretical and empirical findings, collectively underscore a pivotal message: true interpretability is inextricably linked with causality. As NLP systems become part of our daily lives and influence critical decision-making processes in sectors (like law, healthcare, politics, and education), the imperative for faithful explanations has never been greater. This realization calls for a paradigm shift towards exploring and developing more causal-inspired explanation methods.

Understanding when LLM-generated counterfactuals fail. Our empirical results demonstrate that the best method for explaining the causal effect of high-level concepts on model prediction is by employing multiple (Top- K) LLM-generated CFs. However, this does not mean that the problem of interpretability has been solved. There is an impending need for the community to design more challenging benchmarks to identify the areas where the LLMs fall short, misunderstand causality, and fail to provide correct CF explanations (for example, see Kiciman et al. (2023)). Moreover, as we emphasize throughout the paper, utilizing an LLM at inference time is frequently impractical because of its prolonged latency, high financial costs, or privacy concerns, which leads us to the next point.

Closing the gap of efficient explanation methods. As the pursuit of understanding model behavior intensifies, it is paramount that we do not trade off efficiency for quality. The noticeable performance gap between the two approaches we introduced in the paper (the generative and the matching approaches) underscores a ripe avenue for research. For example, the candidate set is crucial for the success of the matching approach, and given the right examples within this set, it has the potential to surpass its generative counterpart. Therefore, an interesting line of work could explore techniques for enriching the candidate set. Bridging this gap could lead to methods that encapsulate the best of both worlds: high quality and operational efficiency.

Constructing new CF-based benchmarks. As we shift to causal-inspired explanation methods, how we evaluate them must evolve in tandem. Accordingly, it is imperative to benchmark explanations against the true causal effect. High-quality CFs, which can serve as a proxy for ground truth in this context, are the basis for such evaluation, as exemplified by the CEBaB benchmark (Abraham et al., 2022). Nevertheless, the traditional approach for constructing such benchmarks relies on human experts and is both financially costly, labor-intensive and fraught with inherent difficulties due to the cognitive demands of the task. Our work demonstrates that LLMs can effectively facilitate this process. We encourage the community to craft benchmarks that resonate with real-world scenarios. These LLM-guided benchmarks can serve as catalyzers for appraising efficient methods as we work towards bridging the existing performance gap.

Ethics statement. No human subjects were directly involved in our research. While explanation methods offer deeper transparency of NLP models, researchers and practitioners should remain vigilant and mindful. These methods, if misused, could over-emphasize or amplify biases present in the model or data. Our novel dataset, derived from tweets taken from SemEval16 (Mohammad et al., 2016), could inherently contain societal biases, including, but not limited to, social prejudices or racism. We acknowledge that tweets can sometimes reflect or amplify societal sentiments, both positive and negative. Furthermore, by leveraging GPT-4 to edit these tweets (e.g., changing the gender of the writer), we recognize the potential for introducing additional biases.

Reproducibility. We have made comprehensive efforts to ensure that researchers can effectively reproduce the results presented in our paper: (1) *Theoretical Foundations*: All pertinent theoretical results, encompassing definitions, assumptions, and proofs, are comprehensively detailed in Appendix A. (2) *Prompts*: The specific prompts we employed for generating examples and counter-

factuals are laid out in Appendix D.1. (3) *Code*: The code utilized in our study is accessible as an attachment to our paper, bundled within a ZIP file. The hyperparameters we used are described in D. (4) *Future Code Documentation*: We recognize the importance of a structured and comprehensively documented codebase. To this end, we are preparing an organized and well-documented GitHub repository. A link to this repository will be accessible in the foreseeable future. (5) *Data and Model Access*: We plan to upload our datasets to the HuggingFace hub, including the new dataset we curated and all the model-generated counterfactuals. In addition, the models we interpreted within the paper will also be hosted.

REFERENCES

- Eldar David Abraham, Karel D’Oosterlinck, Amir Feder, Yair Ori Gat, Atticus Geiger, Christopher Potts, Roi Reichart, and Zhengxuan Wu. Cebab: Estimating the causal effects of real-world concepts on NLP model behavior. In *NeurIPS*, 2022. URL http://papers.nips.cc/paper_files/paper/2022/hash/701ec28790b29a5bc33832b7bdc4c3b6-Abstract-Conference.html.
- Julius Adebayo, Justin Gilmer, Michael Muelly, Ian J. Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In Samy Bengio, Hanna M. Wallach, Hugo Larochelle, Kristen Grauman, Nicolò Cesa-Bianchi, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, pp. 9525–9536, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html>.
- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete Problems in AI Safety. *arXiv:1606.06565 [cs]*, 2016. URL <http://arxiv.org/abs/1606.06565>.
- Maria Antoniak and David Mimno. Bad seeds: Evaluating lexical methods for bias measurement. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 1889–1904, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.148. URL <https://aclanthology.org/2021.acl-long.148>.
- Alexander Balke and Judea Pearl. Probabilistic evaluation of counterfactual queries. In Barbara Hayes-Roth and Richard E. Korf (eds.), *Proceedings of the 12th National Conference on Artificial Intelligence, Seattle, WA, USA, July 31 - August 4, 1994, Volume 1*, pp. 230–237. AAAI Press / The MIT Press, 1994. URL <http://www.aaai.org/Library/AAAI/1994/aaai94-035.php>.
- Eyal Ben-David, Nadav Oved, and Roi Reichart. PADA: example-based prompt learning for on-the-fly adaptation to unseen domains. *Trans. Assoc. Comput. Linguistics*, 10:414–433, 2022. doi: 10.1162/tac1_a_00468. URL https://doi.org/10.1162/tac1_a_00468.
- Umberto Benedetto, Stuart J Head, Gianni D Angelini, and Eugene H Blackstone. Statistical primer: propensity score matching and its alternatives. *European Journal of Cardio-Thoracic Surgery*, 53(6):1112–1117, 2018.
- Alexander Binder, Grégoire Montavon, Sebastian Lapuschkin, Klaus-Robert Müller, and Wojciech Samek. Layer-wise relevance propagation for neural networks with local renormalization layers. In *International Conference on Artificial Neural Networks*, pp. 63–71. Springer, 2016.
- Nitay Calderon, Eyal Ben-David, Amir Feder, and Roi Reichart. Docogen: Domain counterfactual generation for low resource domain adaptation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7727–7746, 2022.
- Nitay Calderon, Subhabrata Mukherjee, Roi Reichart, and Amir Kantor. A systematic study of knowledge distillation for natural language generation with pseudo-target training. In Anna Rogers, Jordan L. Boyd-Graber, and Naoaki Okazaki (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2023, Toronto, Canada, July 9-14, 2023*, pp. 14632–14659. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.818. URL <https://doi.org/10.18653/v1/2023.acl-long.818>.

-
- Carlos Cinelli, Andrew Forney, and Judea Pearl. A crash course in good and bad controls. *Sociological Methods & Research*, pp. 00491241221099552, 2022.
- Kevin Clark, Urvashi Khandelwal, Omer Levy, and Christopher D. Manning. What does BERT look at? An analysis of BERT’s attention. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 276–286, Florence, Italy, August 2019. Association for Computational Linguistics. URL <https://www.aclweb.org/anthology/W19-4828>.
- Alexis Conneau, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. What you can cram into a single $\&\!#\&$ vector: Probing sentence embeddings for linguistic properties. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2126–2136, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1198. URL <https://www.aclweb.org/anthology/P18-1198>.
- Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for evaluating faithfulness of local explanations. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato (eds.), *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pp. 4794–4815. PMLR, 2022. URL <https://proceedings.mlr.press/v162/dasgupta22a.html>.
- Ruofan Deng, Li Pan, and Chloé Clavel. Domain adaptation for stance detection towards unseen target on social media. In *10th International Conference on Affective Computing and Intelligent Interaction, ACII 2022, Nara, Japan, October 18-21, 2022*, pp. 1–8. IEEE, 2022. doi: 10.1109/ACII55700.2022.9953818. URL <https://doi.org/10.1109/ACII55700.2022.9953818>.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Tamar Solorio (eds.), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, pp. 4171–4186. Association for Computational Linguistics, 2019. doi: 10.18653/v1/n19-1423. URL <https://doi.org/10.18653/v1/n19-1423>.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021.
- Amir Feder, Katherine A Keith, Emaad Manzoor, Reid Pryzant, Dhanya Sridhar, Zach Wood-Doughty, Jacob Eisenstein, Justin Grimmer, Roi Reichart, Margaret E Roberts, et al. Causal inference in natural language processing: Estimation, prediction, interpretation and beyond. *arXiv preprint arXiv:2109.00725*, 2021a.
- Amir Feder, Nadav Oved, Uri Shalit, and Roi Reichart. Causalm: Causal model explanation through counterfactual language models. *Computational Linguistics*, 47(2):333–386, 2021b.
- Amir Feder, Nadav Oved, Uri Shalit, and Roi Reichart. CausaLM: Causal model explanation through counterfactual language models. *Computational Linguistics*, 47(2):333–386, June 2021c. doi: 10.1162/coli_a_00404. URL <https://aclanthology.org/2021.cl-2.13>.
- Matthew Finlayson, Aaron Mueller, Sebastian Gehrmann, Stuart Shieber, Tal Linzen, and Yonatan Belinkov. Causal analysis of syntactic agreement mechanisms in neural language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 1828–1843, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.144. URL <https://aclanthology.org/2021.acl-long.144>.
- Matt Gardner, Yoav Artzi, Victoria Basmov, Jonathan Berant, Ben Bogin, Sihao Chen, Pradeep Dasigi, Dheeru Dua, Yanai Elazar, Ananth Gottumukkala, Nitish Gupta, Hannaneh Hajishirzi, Gabriel Ilharco, Daniel Khashabi, Kevin Lin, Jiangming Liu, Nelson F. Liu, Phoebe Mulcaire, Qiang Ning, Sameer Singh, Noah A. Smith, Sanjay Subramanian, Reut Tsarfaty, Eric Wallace, Ally Zhang,

-
- and Ben Zhou. Evaluating models’ local decision boundaries via contrast sets. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 1307–1323, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.findings-emnlp.117. URL <https://aclanthology.org/2020.findings-emnlp.117>.
- Sahaj Garg, Vincent Perot, Nicole Limtiaco, Ankur Taly, Ed H Chi, and Alex Beutel. Counterfactual fairness in text classification through robustness. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 219–226, 2019.
- Itai Gat, Nitay Calderon, Roi Reichart, and Tamir Hazan. A functional information perspective on model interpretation. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato (eds.), *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pp. 7266–7278. PMLR, 2022. URL <https://proceedings.mlr.press/v162/gat22a.html>.
- Atticus Geiger, Kyle Richardson, and Christopher Potts. Neural natural language inference models partially embed theories of lexical entailment and negation. In *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 163–173, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.blackboxnlp-1.16. URL <https://aclanthology.org/2020.blackboxnlp-1.16>.
- Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. *Advances in Neural Information Processing Systems*, 34, 2021.
- Amirata Ghorbani, Abubakar Abid, and James Y. Zou. Interpretation of neural networks is fragile. In *The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019, The Thirty-First Innovative Applications of Artificial Intelligence Conference, IAAI 2019, The Ninth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019, Honolulu, Hawaii, USA, January 27 - February 1, 2019*, pp. 3681–3688. AAAI Press, 2019. doi: 10.1609/aaai.v33i01.33013681. URL <https://doi.org/10.1609/aaai.v33i01.33013681>.
- Bryce Goodman and Seth R. Flaxman. European union regulations on algorithmic decision-making and a ”right to explanation”. *AI Mag.*, 38(3):50–57, 2017. doi: 10.1609/aimag.v38i3.2741. URL <https://doi.org/10.1609/aimag.v38i3.2741>.
- Yash Goyal, Amir Feder, Uri Shalit, and Been Kim. Explaining classifiers with causal concept effect (cace). *CoRR*, abs/1907.07165, 2019. URL <http://arxiv.org/abs/1907.07165>.
- Riccardo Guidotti, Anna Monreale, Salvatore Ruggieri, Franco Turini, Fosca Giannotti, and Dino Pedreschi. A survey of methods for explaining black box models. *ACM Comput. Surv.*, 51(5): 93:1–93:42, 2019. doi: 10.1145/3236009. URL <https://doi.org/10.1145/3236009>.
- Momchil Hardalov, Arnav Arora, Preslav Nakov, and Isabelle Augenstein. Cross-domain label-adaptive stance detection. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*, pp. 9011–9028. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.emnlp-main.710. URL <https://doi.org/10.18653/v1/2021.emnlp-main.710>.
- Momchil Hardalov, Arnav Arora, Preslav Nakov, and Isabelle Augenstein. A survey on stance detection for mis- and disinformation identification. In Marine Carpuat, Marie-Catherine de Marnette, and Iván Vladimir Meza Ruíz (eds.), *Findings of the Association for Computational Linguistics: NAACL 2022, Seattle, WA, United States, July 10-15, 2022*, pp. 1259–1277. Association for Computational Linguistics, 2022. doi: 10.18653/v1/2022.findings-naacl.94. URL <https://doi.org/10.18653/v1/2022.findings-naacl.94>.
- Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4198–4205, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.386. URL <https://aclanthology.org/2020.acl-main.386>.

-
- Sarthak Jain and Byron C Wallace. Attention is not explanation. *arXiv preprint arXiv:1902.10186*, 2019.
- Rohan Jha, Charles Lovering, and Ellie Pavlick. Does data augmentation improve generalization in nlp? *arXiv preprint arXiv:2004.15012*, 2020.
- Divyansh Kaushik, Eduard Hovy, and Zachary C Lipton. Learning the difference that makes a difference with counterfactually-augmented data. *arXiv preprint arXiv:1909.12434*, 2019.
- Divyansh Kaushik, Eduard Hovy, and Zachary C. Lipton. Learning the Difference that Makes a Difference with Counterfactually-Augmented Data. *arXiv:1909.12434 [cs, stat]*, February 2020a. URL <http://arxiv.org/abs/1909.12434>. arXiv: 1909.12434.
- Divyansh Kaushik, Amrith Setlur, Eduard Hovy, and Zachary C Lipton. Explaining the efficacy of counterfactually-augmented data. *arXiv preprint arXiv:2010.02114*, 2020b.
- Emre Kiciman, Robert Ness, Amit Sharma, and Chenhao Tan. Causal reasoning and large language models: Opening a new frontier for causality. *CoRR*, abs/2305.00050, 2023. doi: 10.48550/arXiv.2305.00050. URL <https://doi.org/10.48550/arXiv.2305.00050>.
- Pieter-Jan Kindermans, Kristof T. Schütt, Maximilian Alber, Klaus-Robert Müller, Dumitru Erhan, Been Kim, and Sven Dähne. Learning how to explain neural networks: Patternnet and patternattribution. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018. URL <https://openreview.net/forum?id=Hkn7CBaTW>.
- Dilek Küçük and Fazli Can. Stance detection: A survey. *ACM Comput. Surv.*, 53(1):12:1–12:37, 2021. doi: 10.1145/3369026. URL <https://doi.org/10.1145/3369026>.
- Sören R Künnel, Jasjeet S Sekhon, Peter J Bickel, and Bin Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the national academy of sciences*, 116(10):4156–4165, 2019.
- Jierui Li, Lemao Liu, Huayang Li, Guanlin Li, Guoping Huang, and Shuming Shi. Evaluating explanation methods for neural machine translation. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel R. Tetreault (eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, pp. 365–375. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.35. URL <https://doi.org/10.18653/v1/2020.acl-main.35>.
- Yibing Liu, Haoliang Li, Yangyang Guo, Chenqi Kong, Jing Li, and Shiqi Wang. Rethinking attention-model explainability through faithfulness violation test. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato (eds.), *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pp. 13807–13824. PMLR, 2022. URL <https://proceedings.mlr.press/v162/liu22i.html>.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized BERT pretraining approach. *CoRR*, abs/1907.11692, 2019. URL <http://arxiv.org/abs/1907.11692>.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS’17*, pp. 4768–4777, Red Hook, NY, USA, December 2017. Curran Associates Inc. ISBN 978-1-5108-6096-4.
- Qing Lyu, Marianna Apidianaki, and Chris Callison-Burch. Towards faithful model explanation in NLP: A survey. *CoRR*, abs/2209.11326, 2022. doi: 10.48550/arXiv.2209.11326. URL <https://doi.org/10.48550/arXiv.2209.11326>.
- Christopher D. Manning, Kevin Clark, John Hewitt, Urvashi Khandelwal, and Omer Levy. Emergent linguistic structure in artificial neural networks trained by self-supervision. *Proceedings of the National Academy of Sciences*, 117(48):30046–30054, 2020. ISSN 0027-8424. doi: 10.1073/pnas.1907367117. URL <https://www.pnas.org/content/117/48/30046>.

-
- Chengzhi Mao, Augustine Cha, Amogh Gupta, Hao Wang, Junfeng Yang, and Carl Vondrick. Generative interventions for causal learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3947–3956, 2021.
- Saif M. Mohammad, Svetlana Kiritchenko, Parinaz Sobhani, Xiao-Dan Zhu, and Colin Cherry. Semeval-2016 task 6: Detecting stance in tweets. In Steven Bethard, Daniel M. Cer, Marine Carpuat, David Jurgens, Preslav Nakov, and Torsten Zesch (eds.), *Proceedings of the 10th International Workshop on Semantic Evaluation, SemEval@NAACL-HLT 2016, San Diego, CA, USA, June 16-17, 2016*, pp. 31–41. The Association for Computer Linguistics, 2016. doi: 10.18653/v1/s16-1003. URL <https://doi.org/10.18653/v1/s16-1003>.
- Christoph Molnar. *Interpretable machine learning*. Lulu. com, 2020.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *NeurIPS*, 2022. URL http://papers.nips.cc/paper_files/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract-Conference.html.
- Judea Pearl. *Causality*. Cambridge university press, 2009.
- Danish Pruthi, Rachit Bansal, Bhuwan Dhingra, Livio Baldini Soares, Michael Collins, Zachary C. Lipton, Graham Neubig, and William W. Cohen. Evaluating explanations: How much do explanations from the teacher aid students? *Trans. Assoc. Comput. Linguistics*, 10:359–375, 2022. doi: 10.1162/tacl_a_00465. URL https://doi.org/10.1162/tacl_a_00465.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *CoRR*, abs/1910.10683, 2019. URL <http://arxiv.org/abs/1910.10683>.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pp. 3980–3990. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1410. URL <https://doi.org/10.18653/v1/D19-1410>.
- Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, San Francisco California USA, August 2016. ACM. ISBN 978-1-4503-4232-2. doi: 10.1145/2939672.2939778. URL <https://dl.acm.org/doi/10.1145/2939672.2939778>.
- Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4902–4912, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.442. URL <https://www.aclweb.org/anthology/2020.acl-main.442>.
- Parker Riley, Noah Constant, Mandy Guo, Girish Kumar, David Uthus, and Zarana Parekh. Textsettr: Label-free text style extraction and tunable targeted restyling. *arXiv preprint arXiv:2010.03802*, 2020.
- Daniel Rosenberg, Itai Gat, Amir Feder, and Roi Reichart. Are vqa systems rad? measuring robustness to augmented data with focused interventions. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pp. 61–70, 2021.
- Alexis Ross, Tongshuang Wu, Hao Peng, Matthew E Peters, and Matt Gardner. Tailor: Generating and perturbing text with semantic controls. *arXiv preprint arXiv:2107.07150*, 2021.

-
- Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.*, 1(5):206–215, 2019. doi: 10.1038/s42256-019-0048-x. URL <https://doi.org/10.1038/s42256-019-0048-x>.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of BERT: smaller, faster, cheaper and lighter. *CoRR*, abs/1910.01108, 2019. URL <http://arxiv.org/abs/1910.01108>.
- Naomi Saphra and Adam Lopez. Understanding learning dynamics of language models with SVCCA. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 3257–3267, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1329. URL <https://www.aclweb.org/anthology/N19-1329>.
- Uri Shalit, Fredrik D. Johansson, and David A. Sontag. Estimating individual treatment effect: generalization bounds and algorithms. In Doina Precup and Yee Whye Teh (eds.), *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, volume 70 of *Proceedings of Machine Learning Research*, pp. 3076–3085. PMLR, 2017. URL <http://proceedings.mlr.press/v70/shalit17a.html>.
- Ravi Shekhar, Sandro Pezzelle, Yauhen Klimovich, Aurélie Herbelot, Moin Nabi, Enver Sangineto, and Raffaella Bernardi. FOIL it! find one mismatch between image and language caption. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 255–265, Vancouver, Canada, July 2017. Association for Computational Linguistics. doi: 10.18653/v1/P17-1024. URL <https://aclanthology.org/P17-1024>.
- Ilya Shpitser and Judea Pearl. Identification of conditional interventional distributions. In *22nd Conference on Uncertainty in Artificial Intelligence, UAI 2006*, pp. 437–444, 2006.
- Ilya Shpitser, Tyler J. VanderWeele, and James M. Robins. On the validity of covariate adjustment for estimating causal effects. In Peter Grünwald and Peter Spirtes (eds.), *UAI 2010, Proceedings of the Twenty-Sixth Conference on Uncertainty in Artificial Intelligence, Catalina Island, CA, USA, July 8-11, 2010*, pp. 527–536. AUAI Press, 2010. URL https://dslpitt.org/uai/displayArticleDetails.jsp?mmnu=1&smnu=2&article_id=2078&proceeding_id=26.
- Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences. In *International conference on machine learning*, pp. 3145–3153. PMLR, 2017.
- Suzanna Sia, Anton Belyy, Amjad Almahairi, Madian Khabsa, Luke Zettlemoyer, and Lambert Mathias. Logical satisfiability of counterfactuals for faithful explanations in NLI. In Brian Williams, Yiling Chen, and Jennifer Neville (eds.), *Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence, IAAI 2023, Thirteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2023, Washington, DC, USA, February 7-14, 2023*, pp. 9837–9845. AAAI Press, 2023. doi: 10.1609/aaai.v37i8.26174. URL <https://doi.org/10.1609/aaai.v37i8.26174>.
- Kaitao Song, Xu Tan, Tao Qin, Jianfeng Lu, and Tie-Yan Liu. MpNet: Masked and permuted pre-training for language understanding. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (eds.), *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/c3a690be93aa602ee2dc0ccab5b7b67e-Abstract.html>.
- Paul Soulos, R. Thomas McCoy, Tal Linzen, and Paul Smolensky. Discovering the compositional structure of vector representations with role learning networks. In *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 238–254, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.blackboxnlp-1.23. URL <https://aclanthology.org/2020.blackboxnlp-1.23>.
- Elizabeth A Stuart. Matching methods for causal inference: A review and a look forward. *Statistical science: a review journal of the Institute of Mathematical Statistics*, 25(1):1, 2010.

-
- Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70, ICML'17*, pp. 3319–3328. JMLR.org, 2017.
- Madhumita Sushil, Simon Suster, Kim Luyckx, and Walter Daelemans. Patient representation learning and interpretable evaluation using clinical notes. *J. Biomed. Informatics*, 84:103–113, 2018. doi: 10.1016/j.jbi.2018.06.016. URL <https://doi.org/10.1016/j.jbi.2018.06.016>.
- Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4593–4601, Florence, Italy, July 2019. Association for Computational Linguistics. URL <https://www.aclweb.org/anthology/P19-1452>.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton-Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Rannan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurélien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models. *CoRR*, abs/2307.09288, 2023. doi: 10.48550/arXiv.2307.09288. URL <https://doi.org/10.48550/arXiv.2307.09288>.
- Stijn Vansteelandt, Maarten Bekaert, and Gerda Claeskens. On model selection and model misspecification in causal inference. *Statistical methods in medical research*, 21(1):7–30, 2012.
- Christina Viehmann, Tilman Beck, Marcus Maurer, Oliver Quiring, and Iryna Gurevych. Investigating opinions on public policies in digital media: Setting up a supervised machine learning tool for stance classification. *Communication Methods and Measures*, 17(2):150–184, 2023.
- Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart M. Shieber. Investigating gender bias in language models using causal mediation analysis. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin (eds.), *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html>.
- Feng Wang and Huaping Liu. Understanding the behaviour of contrastive loss. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2021, virtual, June 19-25, 2021*, pp. 2495–2504. Computer Vision Foundation / IEEE, 2021. doi: 10.1109/CVPR46437.2021.00252. URL https://openaccess.thecvf.com/content/CVPR2021/html/Wang_Understanding_the_Behaviour_of_Contrastive_Loss_CVPR_2021_paper.html.
- Sarah Wiegrefe, Ana Marasovic, and Noah A. Smith. Measuring association between labels and free-text rationales. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*, pp. 10266–10284. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.emnlp-main.804. URL <https://doi.org/10.18653/v1/2021.emnlp-main.804>.
- Tongshuang Wu, Marco Tulio Ribeiro, Jeffrey Heer, and Daniel S Weld. Polyjuice: Automated, general-purpose counterfactual generation. *arXiv preprint arXiv:2101.00288*, 2021.
- Zhengxuan Wu, Karel D’Oosterlinck, Atticus Geiger, Amir Zur, and Christopher Potts. Causal proxy models for concept-based model explanations. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *International*

-
- Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, volume 202 of *Proceedings of Machine Learning Research*, pp. 37313–37334. PMLR, 2023. URL <https://proceedings.mlr.press/v202/wu23b.html>.
- Fan Yin, Zhouxing Shi, Cho-Jui Hsieh, and Kai-Wei Chang. On the sensitivity and stability of model interpretations in NLP. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2022, Dublin, Ireland, May 22-27, 2022, pp. 2631–2647. Association for Computational Linguistics, 2022. doi: 10.18653/v1/2022.acl-long.188. URL <https://doi.org/10.18653/v1/2022.acl-long.188>.
- Matthew D. Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars (eds.), *Computer Vision – ECCV 2014*, pp. 818–833, Cham, 2014. Springer International Publishing.
- Xiaoling Zhou and Ou Wu. Implicit counterfactual data augmentation for deep neural networks. *arXiv preprint arXiv:2304.13431*, 2023.
- Ran Zmigrod, Sabrina J. Mielke, Hanna Wallach, and Ryan Cotterell. Counterfactual data augmentation for mitigating gender stereotypes in languages with rich morphology. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 1651–1661, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1161. URL <https://aclanthology.org/P19-1161>.

A A FAITHFUL EXPLANATION

In this section, we prove our main theorem from §3.1. We start by providing definitions for explanation methods and then continue to define the order-faithfulness property. For simplicity, assume the image of the explained models is $\mathbb{R}$.

Definition 1 (Approximated Counterfactual). *Given a DGP $\mathcal{G}$, an explained model f , an example $\mathbf{x}$ whose treatment value is $T(\mathbf{x}) = t$, and an intervention $T : t \rightarrow t'$, we call $\tilde{\mathbf{x}}_{t'}$ **approximated counterfactual** if:*

$$f(\tilde{\mathbf{x}}_{t'}) = \begin{cases} f(\mathbf{x}) & t' = t \\ f(\mathbf{x}_{T=t'}) + \epsilon & \text{otherwise} \end{cases} \quad \mathbb{E}[\epsilon] = 0$$

Where $\mathbf{x}_{T=t'}$ is the golden CF of $\mathbf{x}$ in $\mathcal{G}$ and ϵ is an approximation error.

This definition also suggests why the Top- K technique improves the causal effect estimation: Averaging the prediction of K approximated CFs reduces the variance of the approximation error, making the estimator more robust.

Definition 2 (Approximated Counterfactual Explanation Method). *Given a dataset $\mathbb{D}$ sampled from a DGP $\mathcal{G}$, an explained model f and an intervention $T : t \rightarrow t'$, the **approximated counterfactual explanation method** S_{CF} is defined to be:*

$$S_{CF}(f, T, t, t') = \frac{1}{|\mathbb{D}|} \sum_{\mathbf{x} \in \mathbb{D}} f(\tilde{\mathbf{x}}_{t'}) - f(\tilde{\mathbf{x}}_t)$$

Definition 3 (Non-causal Explanation Method). *Given a dataset $\mathbb{D}$ sampled from a DGP $\mathcal{G}$, an explained model f and an intervention $T : t \rightarrow t'$, let $D[X, T, f]$ be defined to be a set that contains for each unit from $\mathbb{D}$ a triplet of model input, model output, and treatment assignment:*

$$D[X, f, T] = \{(x(\mathbf{u}), f(x(\mathbf{u}), T(\mathbf{u}))) \mid \mathbf{u} \in \mathbb{D}\}$$

An explanation method is called **non-causal explanation method** S_{NC} if it is a function of $D[X, f, T]$:

$$S_{NC}(f, T, t, t') = h(D[X, f, T])$$

For simplicity, we notate the triplet $(x(\mathbf{u}), f(x(\mathbf{u}), T(\mathbf{u})))$ with $(\mathbf{x}, f(\mathbf{x}), T(\mathbf{x}))$. According to Def. 3, the training data of S_{NC} is $D[X, f, T]$, meaning it can be, for example, an unbiased estimator of: $\mathbb{E}_{\mathcal{G}}[f(\mathbf{x})|T = t'] - \mathbb{E}_{\mathcal{G}}[f(\mathbf{x})|T = t]$. However, it may also overlook other concepts than the treatment, such as confounding concepts, which may hint why S_{NC} potentially fails.

Definition 4 (Order-Faithfulness). *Given an i.i.d. dataset $\mathbb{D}$ sampled from a DGP $\mathcal{G}$, an explained model f and a pair of interventions $C_1 : c_1 \rightarrow c'_1, C_2 : c_2 \rightarrow c'_2$, an explanation method S is **order-faithful** if:*

$$\mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S(f, C_1, c_1, c'_1)] > \mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S(f, C_2, c_2, c'_2)] \iff \text{CaCE}_f(C_1, c_1, c'_1) > \text{CaCE}_f(C_2, c_2, c'_2) \quad (13)$$

Lemma 1. *For an explained model f , S_{CF} is order-faithful for any DGP $\mathcal{G}$ and any interventions.*

Proof. We start by connecting the approximated CF to the do operator. From Def. 1, it follows that the expected prediction of an approximated CF is equal to the interventional one:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\tilde{\mathbf{x}}_{t'})] = \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\mathbf{x}_{T=t'}) + \epsilon] = \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\mathbf{x}_{T=t'})] = \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\mathbf{x})|do(T = t')]$$

Combining this result with the Def. 2 and the fact that $P_{\mathbb{D}} = P_{\mathcal{G}}$, we get:

$$\begin{aligned} \mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S_{CF}(f, T, t, t')] &= \mathbb{E}_{\mathbb{D} \sim \mathcal{G}} \left[\frac{1}{|\mathbb{D}|} \sum_{\mathbf{x} \in \mathbb{D}} f(\tilde{\mathbf{x}}_{t'}) - f(\tilde{\mathbf{x}}_t) \right] \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\tilde{\mathbf{x}}_{t'}) - f(\tilde{\mathbf{x}}_t)] \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\tilde{\mathbf{x}}_{t'})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\tilde{\mathbf{x}}_t)] \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\mathbf{x})|do(T = t')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{G}}[f(\mathbf{x})|do(T = t)] \\ &= \text{CaCE}_f(T, t, t') \end{aligned}$$

Thus, S_{CF} is an unbiased estimator of CaCE_f , and it holds that:

$$\mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S_{CF}(f, C_1, c_1, c'_1)] - \mathbb{E}_{\mathbb{D} \sim \mathcal{G}}[S_{CF}(f, C_2, c_2, c'_2)] = \text{CaCE}_f(C_1, c_1, c'_1) - \text{CaCE}_f(C_2, c_2, c'_2)$$

Meaning that S_{CF} is order-faithful for any $\mathcal{G}$ and any interventions. $\square$

Theorem 1. For an explained model f , (1) The approximated CF explanation method S_{CF} is order-faithful for every DGP $\mathcal{G}$ and pair of interventions; (2) For every DGP $\mathcal{G}$, there exist a DGP $\mathcal{G}'$ resulting from a small modification of $\mathcal{G}$, a model f' , and a pair of interventions $C_1 : c_1 \rightarrow c'_1, C_2 : c_2 \rightarrow c'_2$ such that the non-causal explanation method S_{NC} is not order-faithful.

Proof. From Lemma 1 we know that S_{CF} is order-faithful for any DGP and (1) holds.

For part (2) of the theorem, let $\mathcal{G}$ be a DGP with at least two concepts $C_1, \dots, C_K$ (in case $\mathcal{G}$ has only one concept, we can add concept to the DGP and continue with the proof). If S_{NC} is not order-faithful for some two interventions, set $f' = f$ and we are done. Therefore, we assume that S_{NC} is order-faithful for all interventions in $\mathcal{G}$ and w.l.o.g. assume that:

- (a) C_1 has no parent nodes (except exogenous variables). It is guaranteed that at least one concept has no parents since the DGP is a DAG.
- (b) The causal effect of C_1 is bigger than of C_2 : $\text{CaCE}_f(C_1, c_1, c'_1) > \text{CaCE}_f(C_2, c_2, c'_2)$. Otherwise, inverse the corresponding signs of the proof.

Proof sketch of the second part of the theorem: We will construct a new DGP $\mathcal{G}'$ by introducing a new unobserved confounding concept C_0 (this is the “small modification of $\mathcal{G}$ ” from the theorem) and a new explained model f' , such that:

$$\forall i \geq 1 : P_{\mathcal{G}'}(\mathbf{x}, f'(\mathbf{x}), C_i) = P_{\mathcal{G}}(\mathbf{x}, f(\mathbf{x}), C_i) \quad (14)$$

$$\text{CaCE}_{f'}(C_1, c_1, c'_1) < \text{CaCE}_{f'}(C_2, c_2, c'_2) \quad (15)$$

Notice that the first condition Eq. 14 ensures that S_{NC} produces the same explanations also in the new DGP. This is because for an intervention of a concept C_i , S_{NC} is a function of $D[X, f, C_i]$, and the joint distribution from which it is sampled, $P_{\mathcal{G}'}(\mathbf{x}, f'(\mathbf{x}), C_i)$, is the same as in the original DGP $\mathcal{G}$.

Moreover, according to the second condition Eq. 15, for the new $\mathcal{G}'$ and f' , the causal effect of C_1 and C_2 is reversed (compared to the original $\mathcal{G}$ and f). However, S_{NC} produces the same explanations as before (for $\mathcal{G}$ and f), meaning it cannot be order-faithful also for $\mathcal{G}'$ and f' .

We next show that such $\mathcal{G}'$ and f' exist.

Construction: Let f' be a new explained model and $\mathcal{G}'$ be a copy of $\mathcal{G}$ with a new node C_0 , and new edges $(C_0, C_1), (C_0, f'(X))$ (i.e., C_0 is a confounder). C_0 accepts three values: 0, 1 and 2. Let $p = P_{\mathcal{G}}(C_1 = c_1)$ and $p' = P_{\mathcal{G}}(C_1 = c'_1)$. Define the following probabilities:

$$\begin{aligned} P_{\mathcal{G}'}(C_0 = 0) &= p \\ P_{\mathcal{G}'}(C_0 = 1) &= p' \\ P_{\mathcal{G}'}(C_0 = 2) &= 1 - p - p' \\ P_{\mathcal{G}'}(C_1 = c_1 | C_0 = 0) &= P_{\mathcal{G}'}(C_1 = c'_1 | C_0 = 1) = 1 \\ P_{\mathcal{G}'}(C_1 = c_1 | C_0 = 2) &= P_{\mathcal{G}'}(C_1 = c'_1 | C_0 = 2) = 0 \\ \forall c \neq c_1, c'_1 : P_{\mathcal{G}'}(C_1 = c | C_0 = 2) &= \frac{P_{\mathcal{G}}(C_1 = c)}{1 - p - p'} \end{aligned}$$

We conclude that the marginal distribution of C_1 is equal in both DGPs, $P_{\mathcal{G}'}(C_1) = P_{\mathcal{G}}(C_1)$:

$$\begin{aligned} P_{\mathcal{G}'}(C_1 = c_1) &= P_{\mathcal{G}'}(C_1 = c_1 | C_0 = 0) P_{\mathcal{G}'}(C_0 = 0) = p \cdot 1 = P_{\mathcal{G}}(C_1 = c_1) \\ P_{\mathcal{G}'}(C_1 = c'_1) &= P_{\mathcal{G}'}(C_1 = c'_1 | C_0 = 1) P_{\mathcal{G}'}(C_0 = 1) = p' \cdot 1 = P_{\mathcal{G}}(C_1 = c'_1) \\ \forall c \neq c_1, c'_1 : P_{\mathcal{G}'}(C_1 = c) &= P_{\mathcal{G}'}(C_1 = c | C_0 = 2) P_{\mathcal{G}'}(C_0 = 2) \\ &= (1 - p - p') \cdot \frac{P_{\mathcal{G}}(C_1 = c)}{1 - p - p'} = P_{\mathcal{G}}(C_1 = c) \end{aligned}$$

Let d be a difference in causal effects between C_1 and C_2 for f , which is positive following assumption (b) from the beginning of the proof.

$$d = \text{CaCE}_f(C_1, c_1, c'_1) - \text{CaCE}_f(C_2, c_2, c'_2) > 0$$

Let f' be the new explained model. f' has an oracle access¹ to the underlying concepts and is defined to be:

$$f'(\mathbf{x}) = f(\mathbf{x}) - 2d \cdot \mathbb{I}_{[C_1(\mathbf{x})=c_1]} + 2d \cdot \mathbb{I}_{[C_0(\mathbf{x})=0]}$$

Where $\mathbb{I}$ is the indicator function. If $C_0(\mathbf{x}) = 0$ then $C_1(\mathbf{x}) = c_1$ with probability 1 and:

$$f'(\mathbf{x}) = f(\mathbf{x}) - 2d \cdot \mathbb{I}_{[True]} + 2d \cdot \mathbb{I}_{[True]} = f(\mathbf{x}).$$

Conversely, if $C_0(\mathbf{x}) = 1$ then $C_1(\mathbf{x}) = c'_1$ with probability 1 and:

$$f'(\mathbf{x}) = f(\mathbf{x}) - 2d \cdot \mathbb{I}_{[False]} + 2d \cdot \mathbb{I}_{[False]} = f(\mathbf{x})$$

Finally, if $C_0(\mathbf{x}) = 2$ then $C_1(\mathbf{x}) \neq c_1, c'_1$ and:

$$f'(\mathbf{x}) = f(\mathbf{x}) - 2d \cdot \mathbb{I}_{[False]} + 2d \cdot \mathbb{I}_{[False]} = f(\mathbf{x})$$

From the above equations, it is clear that in the DGP $\mathcal{G}'$ without any interventions, for all values of C_0 given $\mathbf{x}$ we have $f'(\mathbf{x}) = f(\mathbf{x})$. Meaning that:

$$P_{\mathcal{G}'}(f'(X)|C_0, X) = P_{\mathcal{G}'}(f'(X)|X) = P_{\mathcal{G}}(f(X)|X)$$

We have shown that $P_{\mathcal{G}'}(C_1) = P_{\mathcal{G}}(C_1)$. Since $\mathcal{G}'$ is a copy of $\mathcal{G}$ (except C_0), and since neither the marginal distribution of the concepts nor the generation process of X did not change, then:

$$\begin{aligned} \forall i = 1, \dots, K : P_{\mathcal{G}}(C_i) &= P_{\mathcal{G}'}(C_i) \\ \forall i = 1, \dots, K : P_{\mathcal{G}}(X|C_i) &= P_{\mathcal{G}'}(X|C_i) \end{aligned}$$

Finally, from definition of f , it depends only on $\mathbf{x}$, thus: $\forall i : P_{\mathcal{G}}(f(X)|X, C_i) = P_{\mathcal{G}}(f(X)|X)$ (the same holds for f' , and we have shown above that $P_{\mathcal{G}'}(f'(X)|C_0, X) = P_{\mathcal{G}'}(f'(X)|X)$). Therefore:

$$\begin{aligned} \forall i = 1, \dots, K : P_{\mathcal{G}}(X, f(X), C_i) &= P_{\mathcal{G}}(C_i)P_{\mathcal{G}}(X|C_i)P_{\mathcal{G}}(f(X)|X, C_i) \\ &= P_{\mathcal{G}}(C_i)P_{\mathcal{G}}(X|C_i)P_{\mathcal{G}}(f(X)|X) \\ &= P_{\mathcal{G}'}(C_i)P_{\mathcal{G}'}(X|C_i)P_{\mathcal{G}'}(f'(X)|X) \\ &= P_{\mathcal{G}'}(C_i)P_{\mathcal{G}'}(X|C_i)P_{\mathcal{G}'}(f'(X)|X, C_i) \\ &= P_{\mathcal{G}'}(X, f'(X), C_i) \end{aligned}$$

Eq. 14 holds and accordingly, $P_{\mathcal{G}'}(D[X, f, C_i]) = P_{\mathcal{G}}(D[X, f, C_i])$, meaning that S_{NC} produces the same explanations for $\mathcal{G}$ and $\mathcal{G}'$. Since S_{NC} is order-faithful in $\mathcal{G}$, and we know from our assumptions on $\mathcal{G}$ that $\text{CaCE}_f(C_1, c_1, c'_1) > \text{CaCE}_f(C_2, c_2, c'_2)$, it follows then that:

$$\mathbb{E}_{\mathbb{D} \sim \mathcal{G}'}[S(f', C_1, c_1, c'_1)] > \mathbb{E}_{\mathbb{D} \sim \mathcal{G}'}[S(f', C_2, c_2, c'_2)]$$

We next show that Eq. 15 holds (reversed causal effect), thus S_{NC} cannot be order-faithful in $\mathcal{G}'$:

$$\begin{aligned} \text{CaCE}_{f'}(C_1, c_1, c'_1) &= \text{CaCE}_f(C_1, c_1, c'_1) - 2d \cdot \text{CaCE}_{\mathbb{I}_{[C_1(\mathbf{x})=c_1]}}(C_1, c_1, c'_1) + 2d \cdot \text{CaCE}_{\mathbb{I}_{[C_0(\mathbf{x})=1]}} \\ &= \text{CaCE}_f(C_1, c_1, c'_1) - 2d \\ &= \text{CaCE}_f(C_1, c_1, c'_1) - (\text{CaCE}_f(C_1, c_1, c'_1) - \text{CaCE}_f(C_2, c_2, c'_2)) - d \\ &= \text{CaCE}_f(C_2, c_2, c'_2) - d < \text{CaCE}_f(C_2, c_2, c'_2) = \text{CaCE}_{f'}(C_2, c_2, c'_2) \end{aligned}$$

The last equation is true as f' is not affected by C_2 in any way except through the f .

□

¹In practice, it can be that the unobserved variable C_0 controls for some modification $\psi(\mathbf{x})$ and the path $X \rightarrow f(X)$ is replaced in the new DGP $\mathcal{G}'$ by $X \rightarrow \psi(X) \rightarrow f'(X)$ and $C_0 \rightarrow \psi(X)$. Nevertheless, S_{NC} has only access to **realizations** of X and f' , and therefore, for simplicity of the notations we assume an oracle.

Table 6: Comparison between the inference time latency (in seconds) of different methods. The top rows present counterfactual generation models, and the middle and bottom tables for matching methods. Notice that all the baselines described in §4.2 have the same latency as the Causal Model and thus are not specified.

Method	Backbone	Explaining a single example			Explaining 100 examples		
		$Top-K = 1$	$Top-K = 10$	$Top-K = 100$	$Top-K = 1$	$Top-K = 10$	$Top-K = 100$
Fine-tune Generative	T5-Base	0.84	1.03	3.09	84	102.2	308.2
Zero-shot Generative	ChatGPT (turbo)	2.45	2.95	4.47	245.3	295.3	447.2
Few-shot Generative	ChatGPT (turbo)	2.52	2.98	4.49	252.3	298.3	449.2
250 candidates		$Top-K = 1$	$Top-K = 10$	$Top-K = 100$	$Top-K = 1$	$Top-K = 10$	$Top-K = 100$
Approx	–	0.86	0.86	0.86	1.95	1.95	1.95
Causal Model	S-Transformer	0.03	0.03	0.03	0.27	0.27	0.27
1000 candidates		$Top-K = 1$	$Top-K = 10$	$Top-K = 100$	$Top-K = 1$	$Top-K = 10$	$Top-K = 100$
Approx	–	0.86	0.86	0.86	1.95	1.95	1.95
Causal Model	S-Transformer	0.03	0.03	0.03	0.27	0.27	0.27

B INFERENCE TIME EFFICIENCY

As discussed in § 1, we seek three attributes in the explanation method: it should be model-agnostic, effective, and efficient. In the main body of this paper, we evaluate the effectiveness of two model-agnostic approaches: Counterfactual generation and matching. We demonstrate the effectiveness and precision of the generative models and our causal matching method compared to other baselines. In this subsection we focus on inference efficiency, an aspect that is vital for providing real-time explanations and explaining vast amounts of data.

To compare the computational efficiency of various explanation methods, we present the inference time latency in Table 6. This latency represents the time (in seconds) required to compute the $\bar{\text{ICaCE}}_f$ for either a single example or a batch of 100 examples. We measure this latency for $Top-K = 1, 10, 100$ using the following methods: (1) The Approx baseline, which employs three fine-tuned RoBERTa models. We first use these models to predict the three confounder concepts of the query examples. Following this, we randomly select matches that correspond to the values of each query example; and (3) Our causal matching method, which first represents the query examples, then computes their cosine similarity with the candidates and finally finds the most similar matches. Notice that other matching baselines have the same latency as our method. All the query and candidate examples are 50 tokens in length, which is also the length of the model-generated counterfactuals. For the matching baselines, we utilized a candidate set comprised of 1000 examples.

As can be seen in Table 2, the latency of the explanation methods we evaluated varies dramatically between them. Interestingly, $Top-K$ does not influence the latency of the matching methods since it merely involves an argmax operation. Similarly, the size of the candidate set has no noticeable impact on latency. Our method also outpaces the Approx baseline, as it utilizes fewer models and avoids the need to filter the candidate set based on the values of the confounder concepts.

The generative baselines are significantly slower than the matching methods, primarily due to the autoregressive token-by-token generation process. For instance, when explaining a single query example, the fine-tuned model is around 30 times slower than matching methods for $K = 1$ and 100 times slower for $K = 100$. In contrast, it is three times faster than zero-shot and few-shot ChatGPT, which has an enormous number of parameters and also suffers from API latency. For batch processing (explaining 100 queries), the benefits of parallelism offered by GPUs make matching methods exceptionally fast. Our method is 1000 times faster than the generative baselines.

In summary, our causal matching method consistently demonstrates superior inference efficiency across various scenarios, making it an ideal choice for applications requiring real-time explanations or the processing of large datasets.

C ADDITIONAL FORMULATIONS

Causal paths. The following three triplets (or X, Y and Z) are the main patterns of a causal graph:

- Chains (or mediators): $X \rightarrow Z \rightarrow Y$. Conditioning on Z blocks the flow from X to Y .

- Forks (or common causes, confounders): $X \leftarrow Z \rightarrow Y$. Conditioning on Z blocks the flow from X to Y .
- Colliders (or common effects): $X \rightarrow Z \leftarrow Y$. The flow from X to Y is blocked by default. However, conditioning on Z opens the flow and induces an association between X and Y .

Back-door criterion (Pearl, 2009). A set of variables $\mathbb{Z}$ satisfies the back-door criterion relative to (X, Y) in a directed acyclic graph $\mathcal{G}$ if:

- No node in $\mathbb{Z}$ is a descendant of X .
- Z blocks every path between X and Y that contains an arrow into X .

The adjustment criterion (Shpitser et al., 2010) was later devised to handle cases in which Z may explicitly contain descendants of X ; however, it is unnecessary for our causal graph described in Figure 1. We term the set $\mathbb{Z}$, which satisfies the back-door criterion as the adjustment set.

C.1 CEBAB - CAUSAL ESTIMATION-BASED BENCHMARK (OPENTABLE REVIEWS)

The adjustment set of CEbaB. In our causal graph (Figure 1), when estimating the causal effect of an aspect, for example, S , on the model f , the following paths should be taken into account (F and N w.l.o.g):

- $S \rightarrow X \rightarrow f(X)$
- $S \leftarrow U \rightarrow F \rightarrow X \rightarrow f(X)$
- $S \leftarrow U \rightarrow F \rightarrow Y \leftarrow N \rightarrow X \rightarrow f(X)$
- $S \rightarrow Y \leftarrow F \rightarrow X \rightarrow f(X)$
- $S \rightarrow Y \leftarrow F \leftarrow U \rightarrow N \rightarrow X \rightarrow f(X)$
- (when $Y \rightarrow X$) $S \rightarrow Y \rightarrow X \rightarrow f(X)$
- (when $Y \leftarrow X$) $S \rightarrow Y \leftarrow X \rightarrow f(X)$

The adjustment set is F, N, A , and Y must not be adjusted. Accordingly, the set of matches $\mathbb{X}_M$ should contain texts with the same aspect values (excluding the treatment) as the query example. Moreover, it clarifies why we use the term misspecified to describe the sets $\mathbb{X}_M$ and $\mathbb{X}_{CF}$. This is because they contain texts that at least one of their aspect values from the query is different or was changed to another.

Increased precision when conditioning on V . The exogenous variable V is a direct cause of the text X that mediates between concepts (e.g., S) and the model prediction $f(X)$. For example, the variable V can account for the syntax, writing style, or length of X . Controlling for V (hypothetically, since it is not observed) will not bias the causal effect estimation (of CaCE, and not of ICaCE, which by definition should control for V) since it does not open any back-door paths from S to Y . On the other hand, controlling for V can increase the estimation precision of CaCE (Cinelli et al., 2022).

Consider the case where the model f learns spurious correlations between V and Y (e.g., the sentiment of long texts tends to be negative). We use the Approx matching technique (which controls all the adjusted variables: F , N , and A and sample a unit from the control group which shares the same adjustment values as the query example) to calculate the $\widehat{\text{CaCE}}_f$ (see Eq.2). Since V is independent of any other variable (except X), the $\widehat{\text{CaCE}}_f$ estimator is not be biased asymptotically. However, when the queries and the candidate sets are finite (and small), the spurious correlations f learned might be amplified by the non-representing sample. Controlling for V (finding matches that also share the same attributes, e.g., writing style) mitigates this bias and increases the precision of the estimation.

Therefore, utilizing counterfactuals while training the causal representation model used for matching is vital. In contrast to $\mathbb{X}_M$, the set of counterfactuals $\mathbb{X}_{CF}$ should share the same values of V , and the causal model learns to consider it when finding a match.

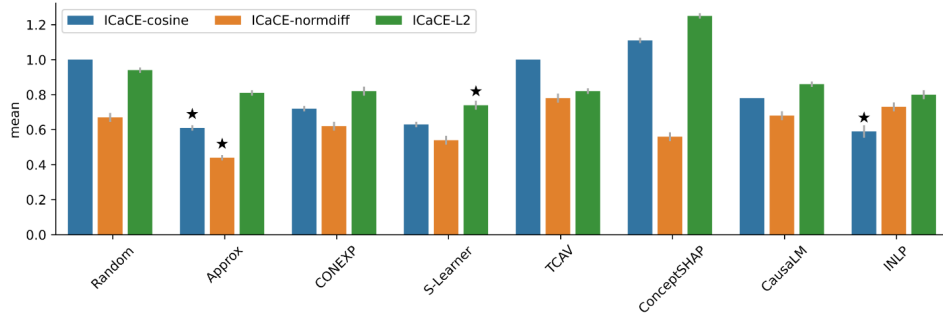


Figure 6: A comparison between different explanation methods adapted from Abraham et al. (2022). The best performing methods are the simple Approx baseline and the S-learner.

NI
Rate the restaurant on a scale from 1 to 5.
review:
[TEXT]

rating:

Figure 7: The prompt we use for rating CEBaB reviews and extracting the five-way sentiment distribution (of the Llamas models). We extract the next token probabilities of 1-5 tokens. [TEXT] is a review.

D IMPLEMENTATION DETAILS

We use $\tau = 0.1$, train the causal representation models for 12 epochs, and select the model checkpoint which minimizes the objective of Eq. 8 (or its modified version for the ablation models). Notice that when using the Err as the selection criterion, one might be able to select a more robust checkpoint. However, this violates the model-agnostic property, as calculating the Err requires access to the explained model. The backbone SentenceTransformer of our model is MPNet (all-mpnet-base-v2) (Song et al., 2020). We use a learning rate of $5e-6$ for training the causal representation models. We use a learning rate of $1e-5$ and a batch size of 16 for the fine-tuned baselines, explained models and concept predictors. The concept predictor models are used for filtering misspecified CFs (for constructing the $\mathbb{X}_{CF}$ and $\mathbb{X}_{\neg CF}$ sets) and for the Approx and Propensity score baselines. We fine-tune a RoBERTa model using the annotated train set for each of the four concepts.

For generating counterfactuals, we use the OpenAI API with the following models: gpt-3.5-turbo (referred to as ChatGPT in the paper) and gpt-4 (GPT-4).

D.1 PROMPTS AND EXAMPLES

NI I'm providing a review from OpenTable and your task is to edit the review according to my instruction.

---- Input Review ----

[TEXT]

----- Instruction -----

The reviewer gave a POSITIVE score to the FOOD aspect. You should edit the review such that there is a NEGATIVE score to the FOOD aspect.

We consider four aspects FOOD, AMBIANCE, SERVICE, NOISE, make sure you change ONLY the rating of FOOD.

----- Edited Review -----

Figure 8: The zero-shot prompt we use for generating CFs for CEBaB examples. [TEXT] is the review. The instruction is changed according to the treatment (e.g., in this prompt, we change the value of the food aspect from positive to negative).

NI I'm providing a review from OpenTable and your task is to edit the review according to my instruction.

Here are a few examples:

----- Example 1 -----

----- Review -----

[TEXT]

----- Instruction -----

The reviewer gave a POSITIVE score to the FOOD aspect. You should edit the review such that there is a NEGATIVE score to the FOOD aspect.

We consider four aspects FOOD, AMBIANCE, SERVICE, NOISE, make sure you change ONLY the rating of FOOD.

----- Edited Review -----

[COUNTERFACTUAL]

----- Example 2 -----

----- Review -----

[TEXT]

----- Instruction -----

The reviewer gave a NEGATIVE score to the FOOD aspect. You should edit the review such that there is NO INFORMATION about the food aspect.

We consider four aspects FOOD, AMBIANCE, SERVICE, NOISE, make sure you change ONLY the rating of FOOD.

----- Edited Review -----

[COUNTERFACTUAL]

----- Example 3 -----

----- Review -----

[TEXT]

----- Instruction -----

The reviewer gave a POSITIVE score to the FOOD aspect. You should edit the review such that there is a NEGATIVE score to the FOOD aspect.

We consider four aspects FOOD, AMBIANCE, SERVICE, NOISE, make sure you change ONLY the rating of FOOD.

----- Edited Review -----

[COUNTERFACTUAL]

----- Input Review ----

[TEXT]

----- Instruction -----

The reviewer gave a POSITIVE score to the FOOD aspect. You should edit the review such that there is a NEGATIVE score to the FOOD aspect.

We consider four aspects FOOD, AMBIANCE, SERVICE, NOISE, make sure you change ONLY the rating of FOOD.

Figure 9: The few-shot prompt we use for generating CFs for CEBaB examples. [TEXT] is a review and [COUNTERFACTUAL] is a human-written CF. Every prompt consists of three demonstrations where the aspect of the treatment is changed. Since there are four aspects, only 12 human-written CFs are required.

NI The following tweet may discuss [DOMAIN]. Please edit the tweet to make it sound as if [DESCRIPTION].
Then, specify if the new tweet favor, oppose, or remain neutral about [DOMAIN].
Respond only with this JSON format: `{"new\_tweet": X, "label": Y}` where Y must be "favor", "oppose" or "neutral".

Tweet:
[TEXT]

Figure 10: The prompt we use for generating examples for the new stance detection setup. Notice - these examples are used for enriching the dataset with more writer profiles. [DOMAIN] is the tweet topic/subject (e.g., abortions), and [TEXT] is the original tweet which we ask the LLM (GPT-4) to edit. The [DESCRIPTION] describes the new profile of the writer. For example, it can be: “an elder (60+) farmer wrote it”. In addition, we ask the LLM to predict the label of the new tweet.

NI The following tweet may discuss [DOMAIN]. The writer's profile is - [DESCRIPTION].
Please edit the tweet to make it sound as if [EDIT_DESCRIPTION].
Then, specify if the new tweet favor, oppose, or remain neutral about [DOMAIN].
Respond only with this JSON format: `{"new\_tweet": X, "label": Y}` where Y must be "favor", "oppose" or "neutral".

Tweet:
[TEXT]

Figure 11: The prompt we use for generating ground-truth CFs for examples from the new stance detection setup. [DOMAIN] is the tweet topic/subject (e.g., abortions), and [TEXT] is the original tweet which we ask the LLM (GPT-4) to edit. The [DESCRIPTION] describes the old profile of the writer (e.g., “Age: Unknown, Gender: Female, Job: professor”). The [EDIT DESCRIPTION] describes the new profile, for example, “an elder (60+) farmer wrote it”, or “the age, gender, or job of the writer is unknown”. In addition, we ask the LLM to predict the label of the new tweet.

NI The following tweet may discuss [DOMAIN]. The writer's profile is - [DESCRIPTION].
Please edit the tweet to make it sound as if [EDIT_DESCRIPTION].
Respond only with the revised text.

Tweet:
[TEXT]

Figure 12: The zero-shot prompt we use for generating CFs for examples from the new stance detection setup. This prompt is used for benchmarking the LLM (ChatGPT), and not for generating instances for the test set. See Figure 11 caption for more details.

Table 7: Fractions in percentages of the variables (joint probability) in the new setup (Stance Detection, §6). For example, according to row **CC** and column **T**, 7.3% of the data are tweets about Climate Change (Subject concept) written by a teenager (Age concept). There are a total of 4000 examples divided into the train ($N = 1000$), matching ($N = 2250$), dev ($N = 250$), and test ($N = 500$) sets. **Orig. Label** is the label of the original tweet before any modifications, and **Label** is the label predicted by GPT-4 after modifying the tweet. Legend: *Subject* – **Ab** (Abortions), **At** (Atheism), **CC** (Climate Change) **Fe** (Feminism) **HC** (Hillary Clinton); *Age* – **?** (Unknown), **T** (Teenager), **E** (Elder); *Job* – **?** (Unknown), ♀ (Female), ♂ (Male); *Job* – **?** (Unknown), **F** (Farmer), **P** (Professor); *Label* – **?** (Neutral), **X** (Oppose), ✓ (Support); *Orig. Label* – **?** (Neutral), **X** (Oppose), ✓ (Support).

	Subject					Age			Gender			Job			Label			Orig. Label		
	Ab	At	CC	Fe	HC	?	T	E	?	♀	♂	?	F	P	?	X	✓	?	X	✓
Ab	20.0	0.0	0.0	0.0	0.0	7.2	3.6	9.1	10.1	3.8	6.2	8.6	7.2	4.1	13.4	5.2	1.4	4.9	11.6	3.5
At	0.0	20.0	0.0	0.0	0.0	7.7	5.4	7.0	9.7	4.2	6.0	9.7	6.5	3.8	13.1	5.6	1.3	4.0	12.4	3.6
CC	0.0	0.0	20.0	0.0	0.0	8.7	7.3	4.0	9.5	6.2	4.3	8.4	4.0	7.6	12.5	1.0	6.5	7.6	0.8	11.5
Fe	0.0	0.0	0.0	20.0	0.0	9.2	5.4	5.4	8.3	4.6	7.2	8.4	6.6	4.9	13.1	1.8	5.0	3.4	10.8	5.8
HC	0.0	0.0	0.0	0.0	20.0	8.2	4.5	7.3	9.4	4.1	6.5	8.0	8.4	3.6	12.7	5.3	2.0	5.5	11.2	3.3
?	7.2	7.7	8.7	9.2	8.2	41.0	0.0	0.0	11.2	13.0	16.8	8.4	18.1	14.6	28.2	6.5	6.3	11.0	18.9	11.1
T	3.6	5.4	7.3	5.4	4.5	0.0	26.2	0.0	14.9	6.2	5.2	16.6	4.2	5.4	16.5	2.6	7.1	8.7	4.9	12.6
E	9.1	7.0	4.0	5.4	7.3	0.0	0.0	32.8	20.9	3.6	8.2	18.2	10.4	4.1	20.1	9.9	2.8	5.8	23.0	4.0
♀	10.1	9.7	9.5	8.3	9.4	11.2	14.9	20.9	47.0	0.0	0.0	11.6	21.2	14.1	30.4	9.8	6.8	11.4	23.3	12.3
♂	3.8	4.2	6.2	4.6	4.1	13.0	6.2	3.6	0.0	22.9	0.0	14.4	3.4	5.1	14.0	2.6	6.3	6.1	6.4	10.4
♂	6.2	6.0	4.3	7.2	6.5	16.8	5.2	8.2	0.0	0.0	30.1	17.2	8.2	4.8	20.6	6.5	3.1	8.0	17.2	5.0
?	8.6	9.7	8.4	8.4	8.0	8.4	16.6	18.2	11.6	14.4	17.2	43.2	0.0	0.0	26.0	9.1	8.2	11.4	19.0	12.7
F	7.2	6.5	4.0	6.6	8.4	18.1	4.2	10.4	21.2	3.4	8.2	0.0	32.8	0.0	22.1	8.2	2.5	8.5	21.8	2.5
P	4.1	3.8	7.6	4.9	3.6	14.6	5.4	4.1	14.1	5.1	4.8	0.0	0.0	24.0	16.8	1.6	5.6	5.5	6.0	12.5
?	13.4	13.1	12.5	13.1	12.7	28.2	16.5	20.1	30.4	14.0	20.6	26.0	22.1	16.8	64.8	0.0	0.0	24.3	27.2	13.3
X	5.2	5.6	1.0	1.8	5.3	6.5	2.6	9.9	9.8	2.6	6.5	9.1	8.2	1.6	0.0	19.0	0.0	0.5	17.4	1.1
✓	1.4	1.3	6.5	5.0	2.0	6.3	7.1	2.8	6.8	6.3	3.1	8.2	2.5	5.6	0.0	0.0	16.2	0.7	2.2	13.3
?	4.9	4.0	7.6	3.4	5.5	11.0	8.7	5.8	11.4	6.1	8.0	11.4	8.5	5.5	24.3	0.5	0.7	25.5	0.0	0.0
X	11.6	12.4	0.8	10.8	11.2	18.9	4.9	23.0	23.3	6.4	17.2	19.0	21.8	6.0	27.2	17.4	2.2	0.0	46.8	0.0
✓	3.5	3.6	11.5	5.8	3.3	11.1	12.6	4.0	12.3	10.4	5.0	12.7	2.5	12.5	13.3	1.1	13.3	0.0	0.0	27.7